

Categorical perception as probabilistic inference

Computational modeling of language
Spring 2026

Bureaucracy

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- Our last homework will be issued this week, due next week.

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- Next week:
 - Tue: A research lecture by me
 - Thu: Review, initial prep for final exam

Probabilistic models of cognition

Probabilistic models of cognition

Perception and cognition as inference

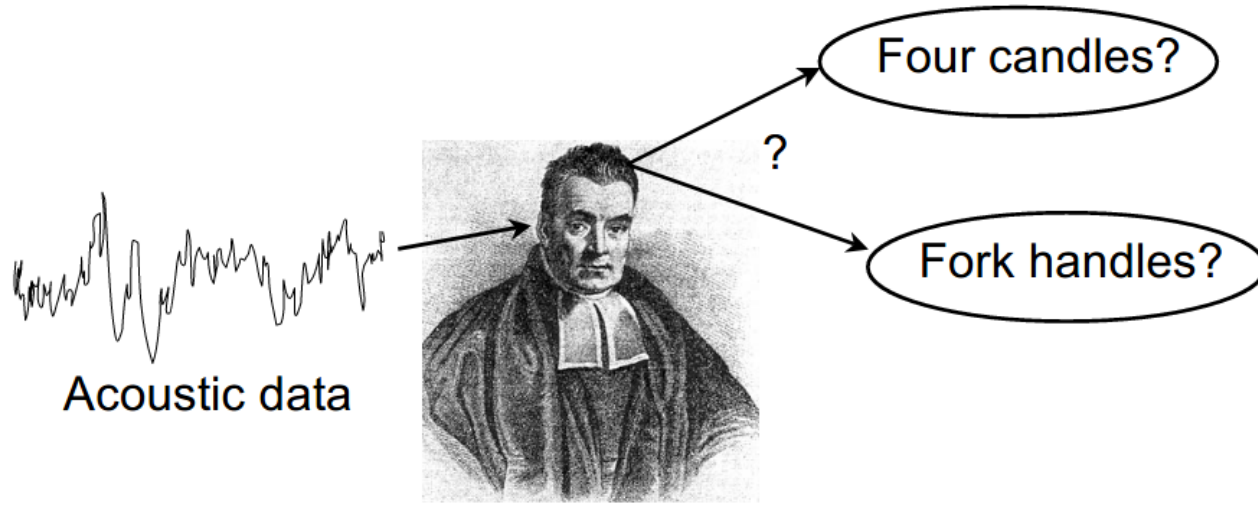
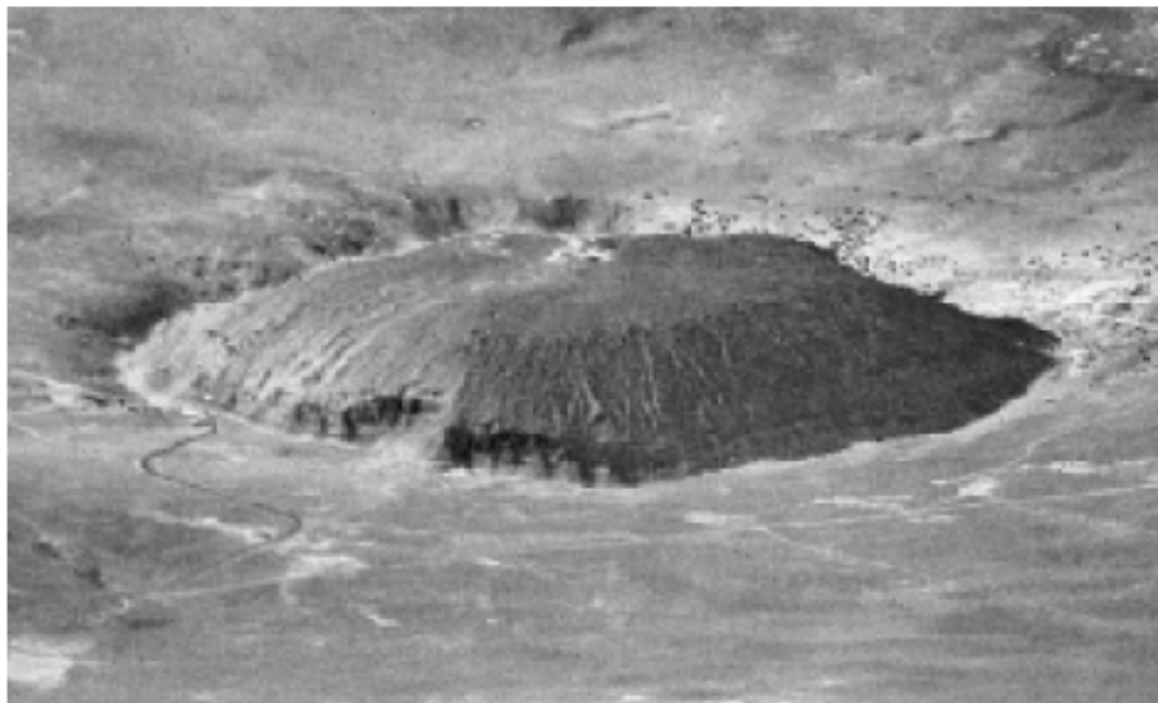


Figure 1.5.: Thomas Bayes trying to make sense of a London accent, which removes the *h* sound from the word *handle*, so the phrase *fork handles* is pronounced *fork 'andles*, and therefore sounds like *four candles* (see Fork Handles YouTube clip by The Two Ronnies).





Seeing is not a direct apprehension of reality,
as we often like to pretend. Quite the
contrary: **seeing is inference from
incomplete information...**

E.T. Jaynes 2003

Probabilistic inference is often taken as a **normative standard** against which human performance can be compared.

Categorical perception

Categorical perception

Vowel perception

The Influence of Categories on Perception: Explaining the Perceptual Magnet Effect as Optimal Statistical Inference

Naomi H. Feldman
Brown University

Thomas L. Griffiths
University of California, Berkeley

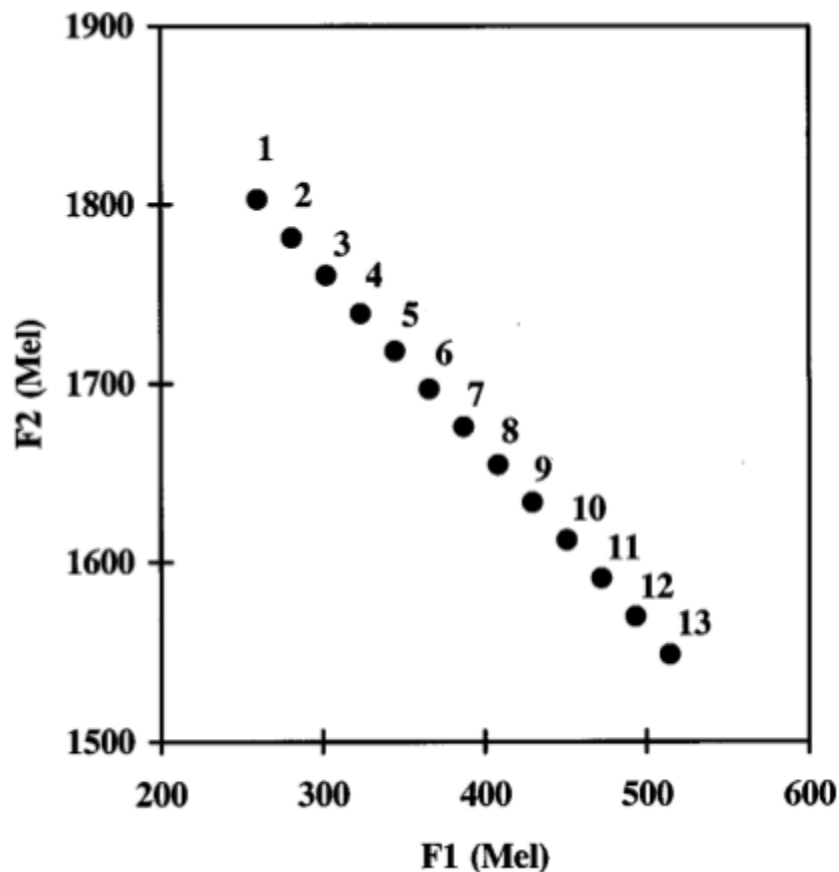
James L. Morgan
Brown University

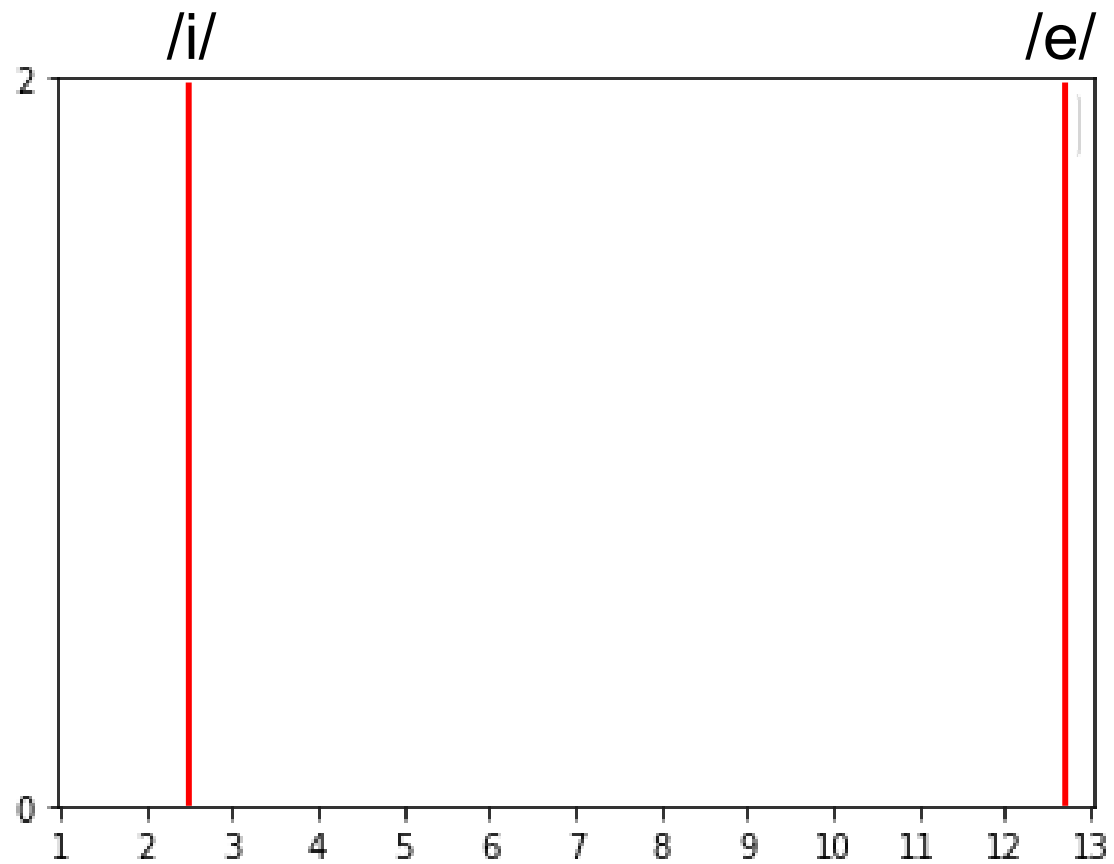
The perceptual magnet effect

A form of **categorical perception**:

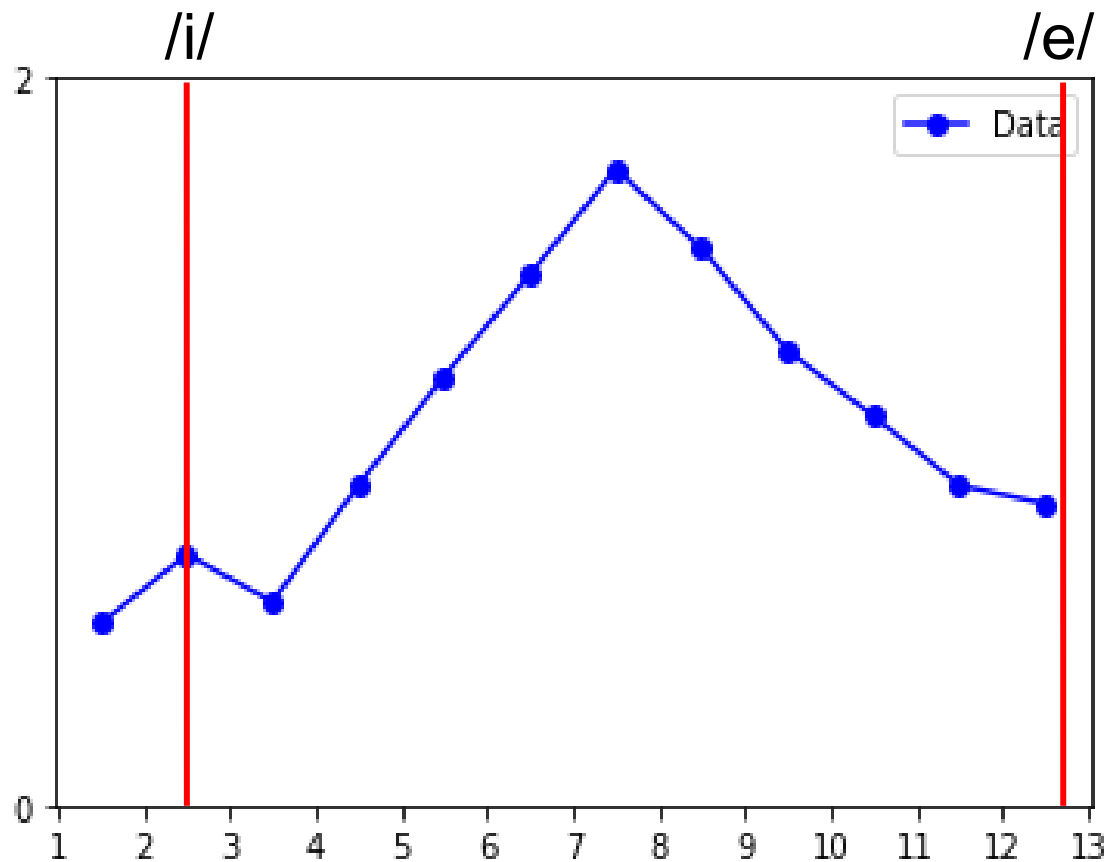
- “discriminability between vowels is reduced near prototypical vowel sounds” (Feldman et al.)

A **vowel continuum**
from /i/ to /e/, evenly
spaced along mel
scale.





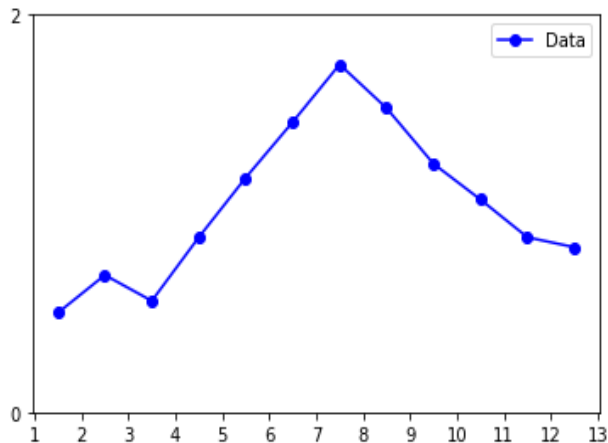
Relative perceptual distance
between neighboring stimuli



Iverson & Kuhl 1995

The perceptual magnet effect

- “discriminability between vowels is reduced near prototypical vowel sounds”

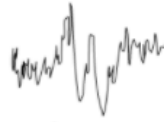


perceptual warping

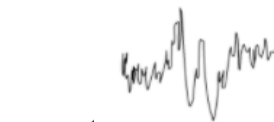
Central claim of Feldman et al. 2009

The perceptual magnet effect is a natural consequence of “**optimally solving the statistical problem of perception in noise**”.

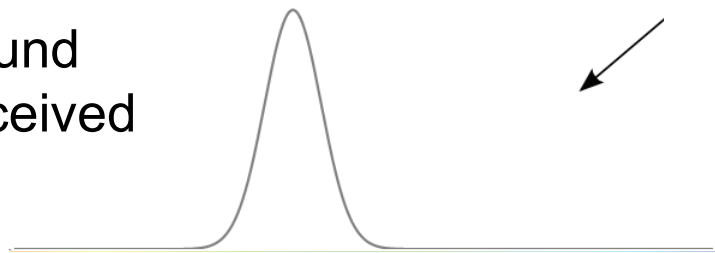
sound produced



sound produced



sound
received

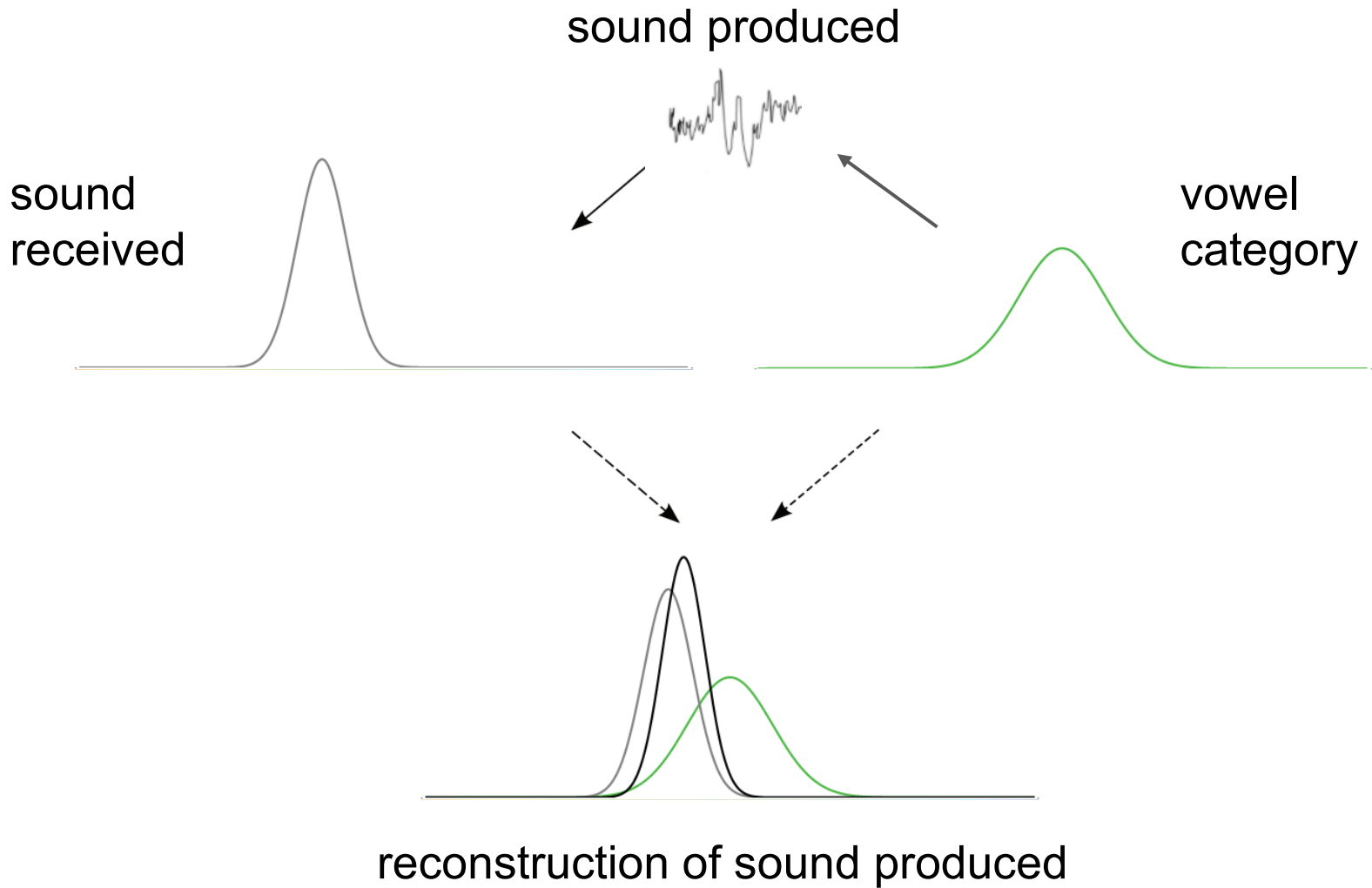


sound produced

sound
received

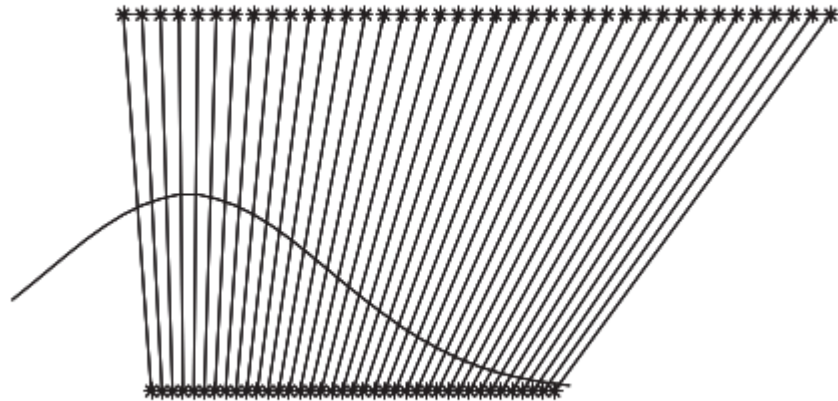
vowel
category





(a)

Actual Stimulus

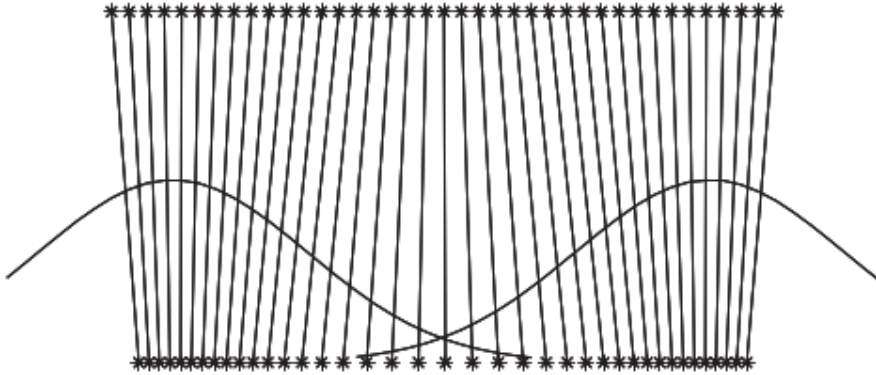


Perceived Stimulus

Bias toward category
prototype

(b)

Actual Stimulus

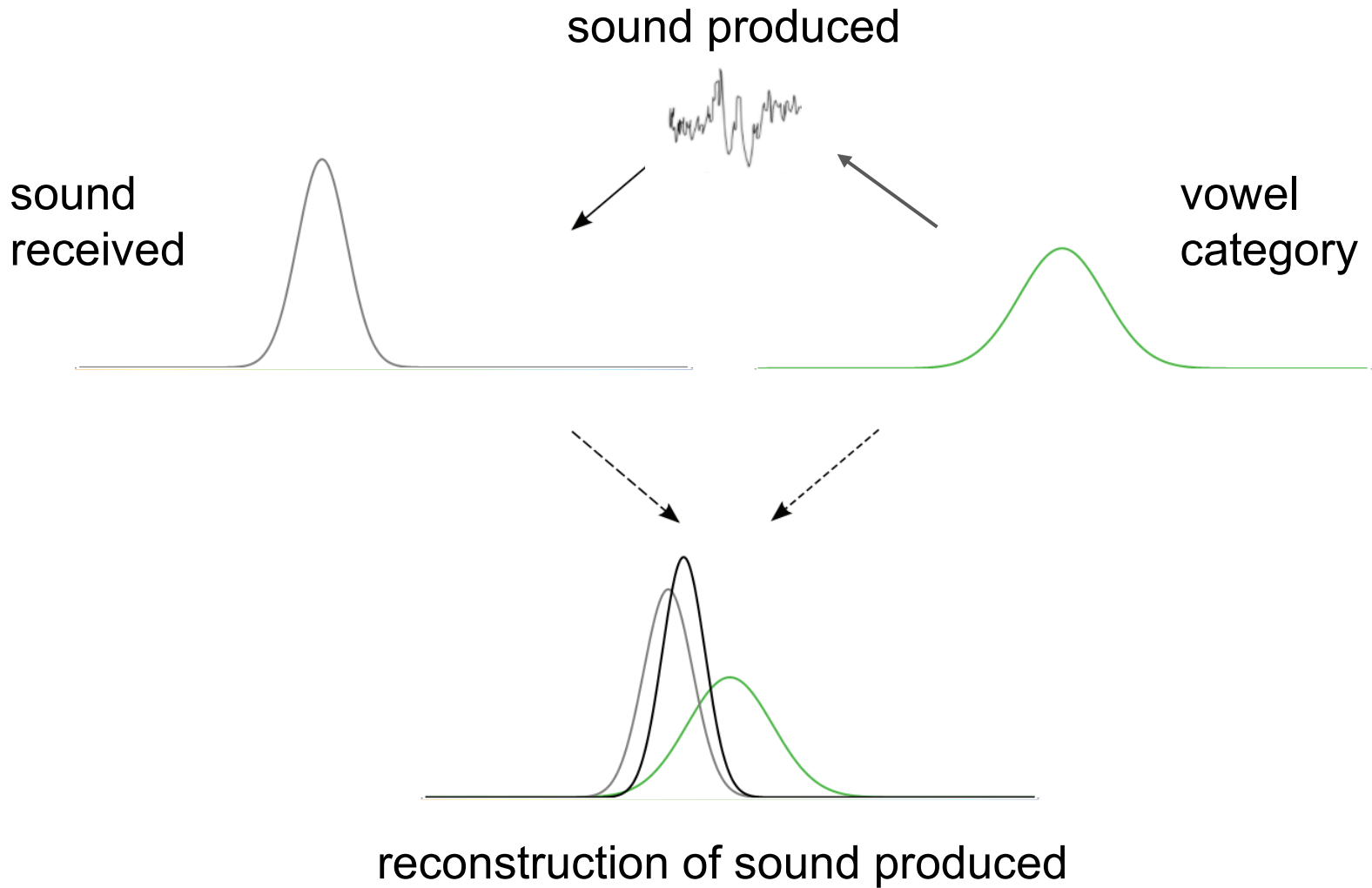


Perceived Stimulus

Bias toward category
prototype

⇒

poorer discrimination
near prototype



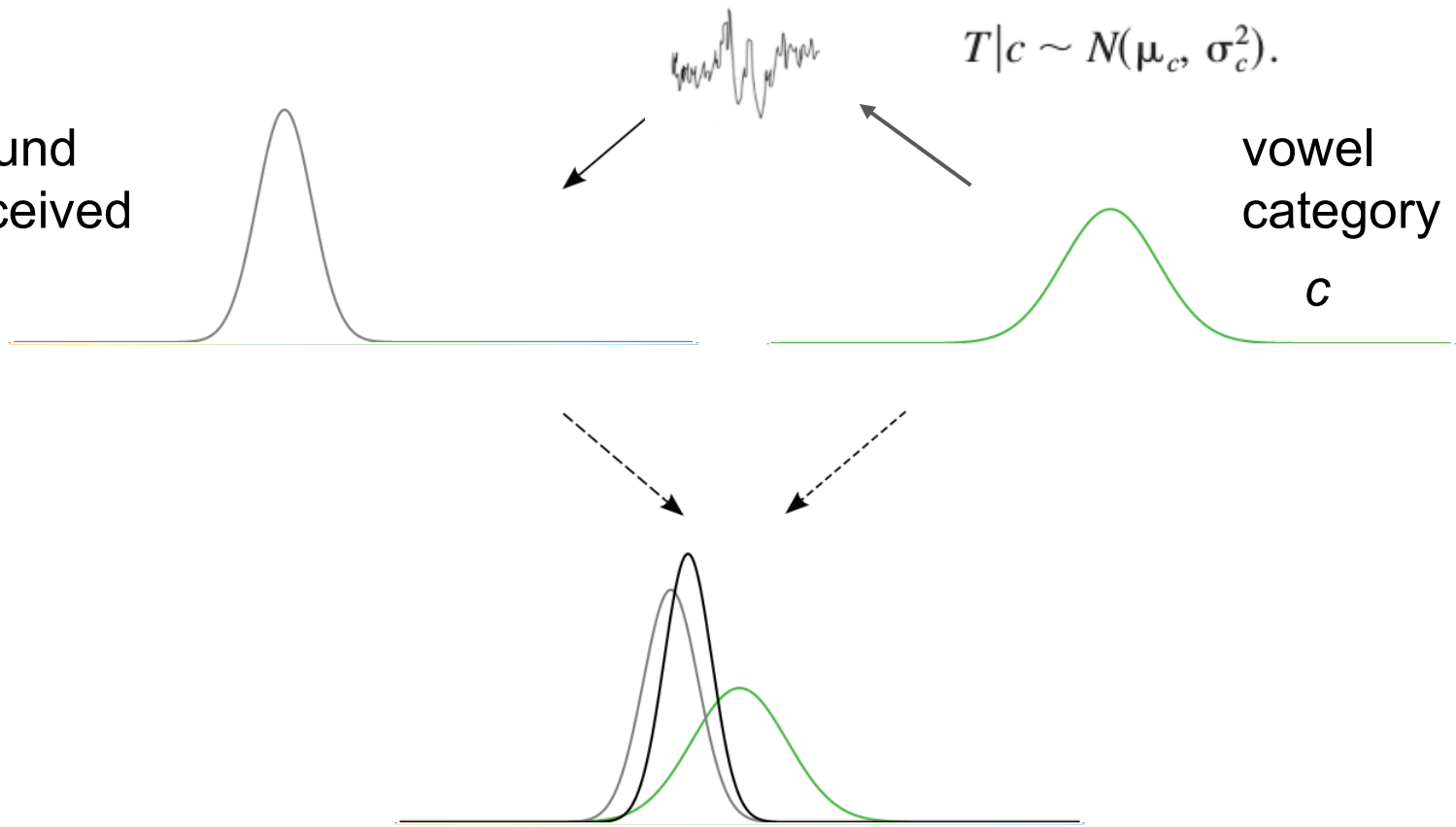
sound produced T

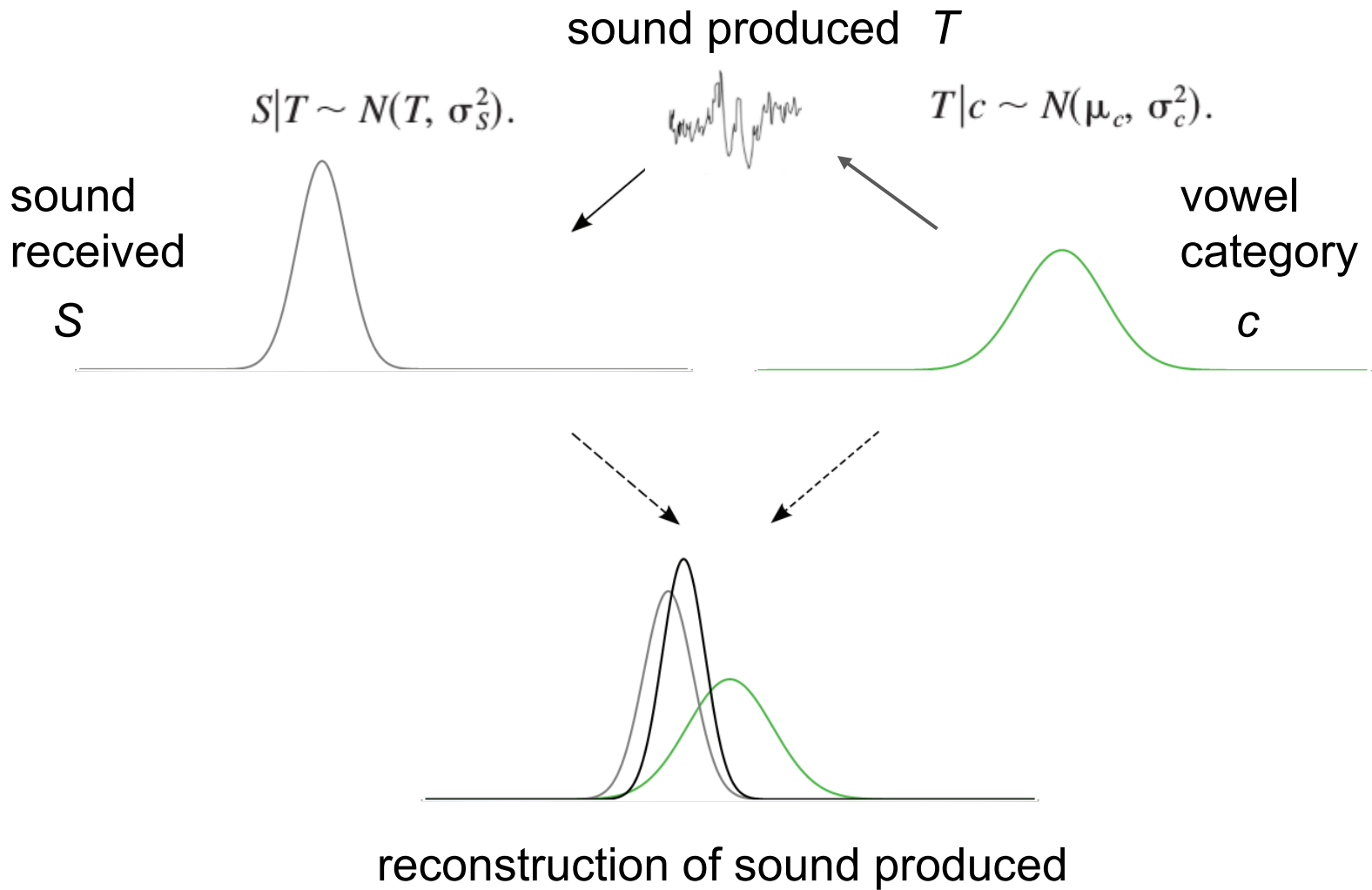
$$T|c \sim N(\mu_c, \sigma_c^2).$$

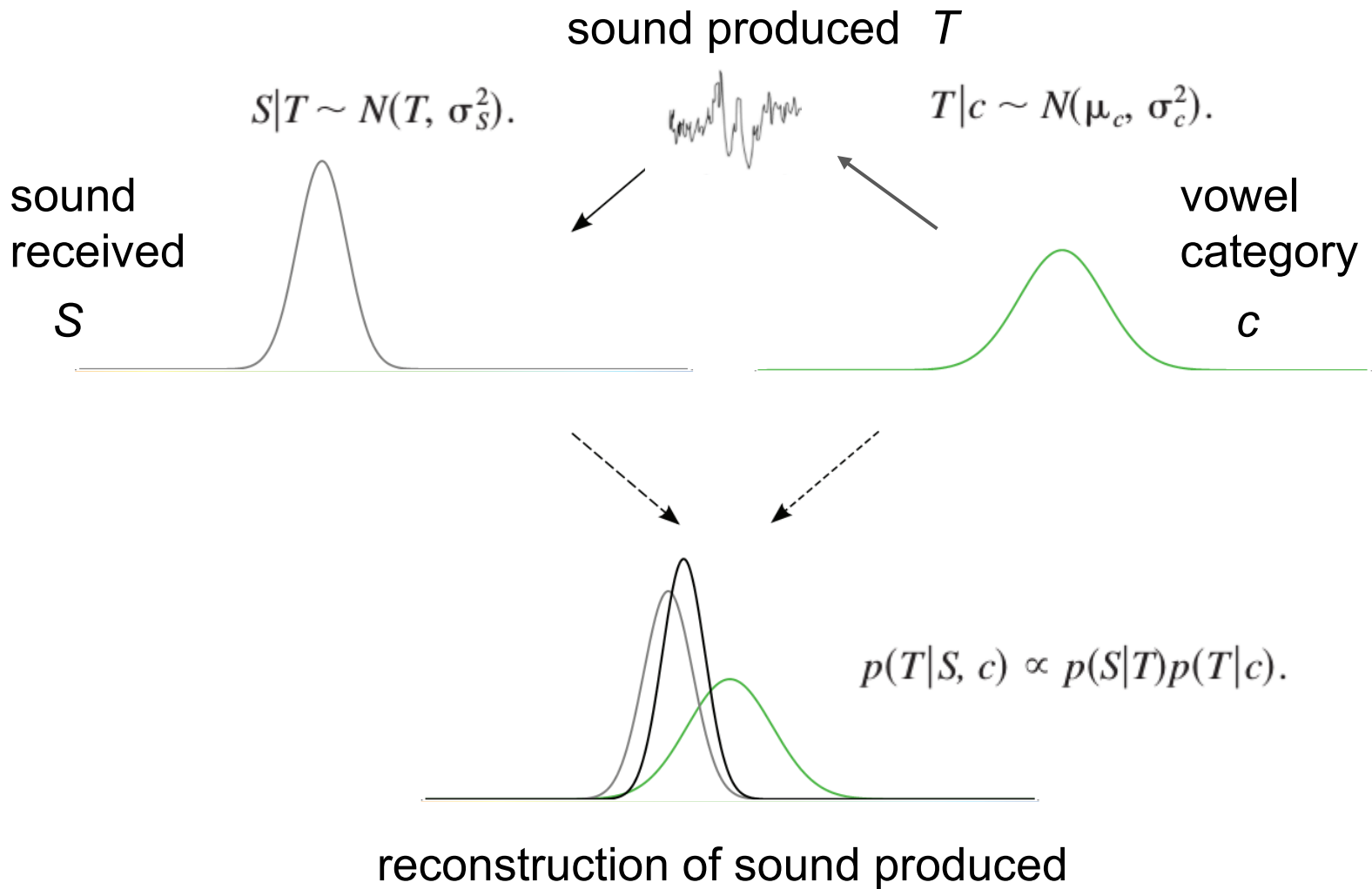
sound
received

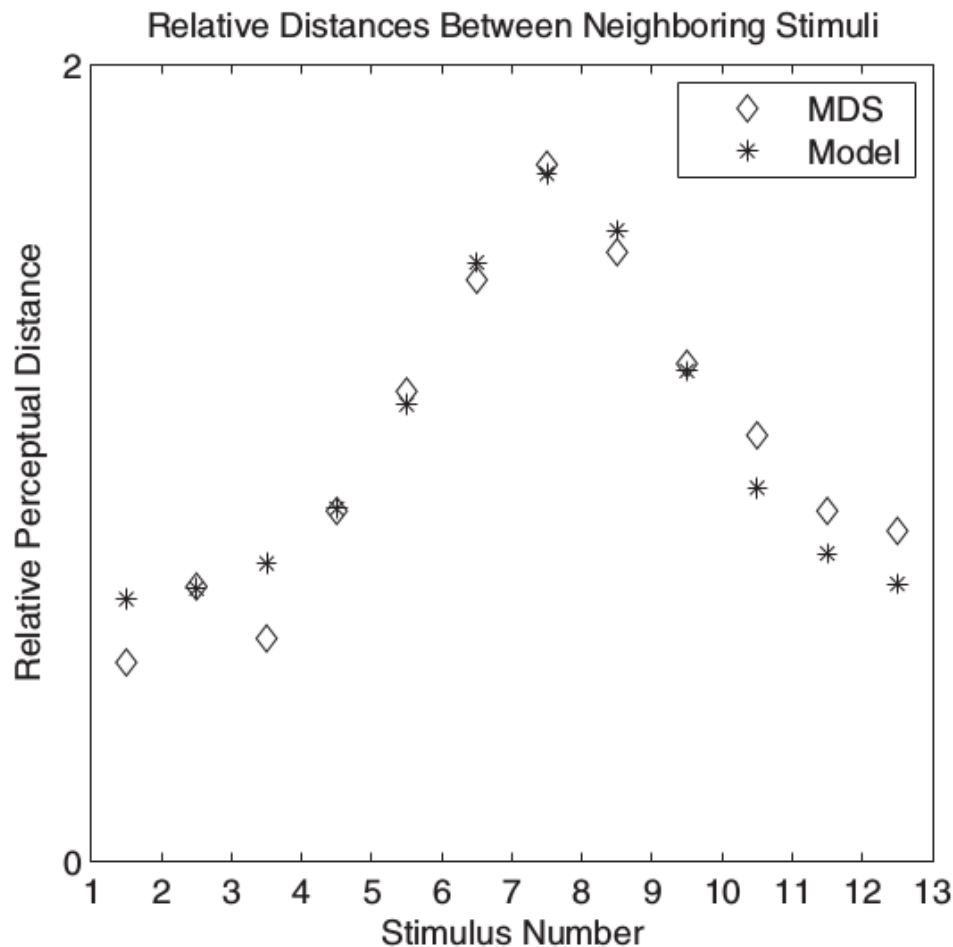
vowel
category
 c

reconstruction of sound produced



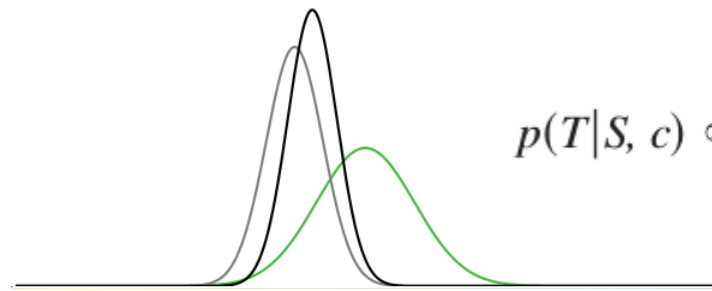






Model fit to Iverson & Kuhl 1995 data that we saw earlier.

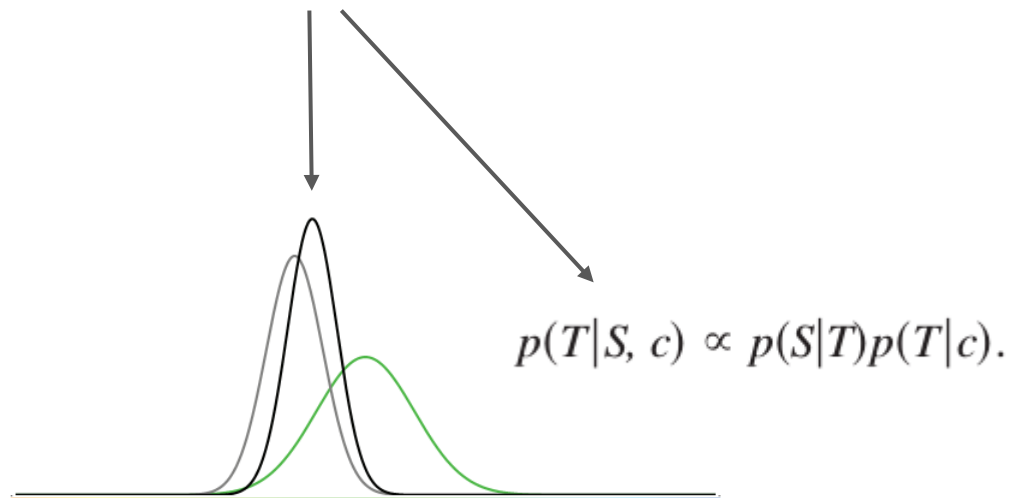
2 categories, with means and variances determined independently.



$$p(T|S, c) \propto p(S|T)p(T|c).$$

reconstruction of sound produced

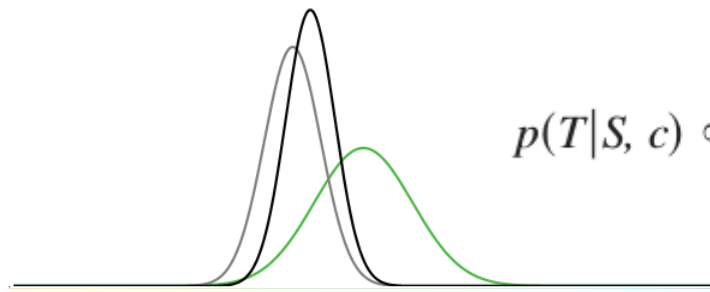
Posterior distribution



reconstruction of sound produced

Analytic form for mean of posterior distribution,
for two-category model:

$$E[T|S] = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_s^2} S + \frac{\sigma_s^2}{\sigma_c^2 + \sigma_s^2} \sum_c p(c|S) \mu_c$$



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reconstruction of sound produced

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- **2 terms:** **stimulus**, and **category means**

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- 2 terms: **stimulus**, and **category means**
- Weighted by **certainty**

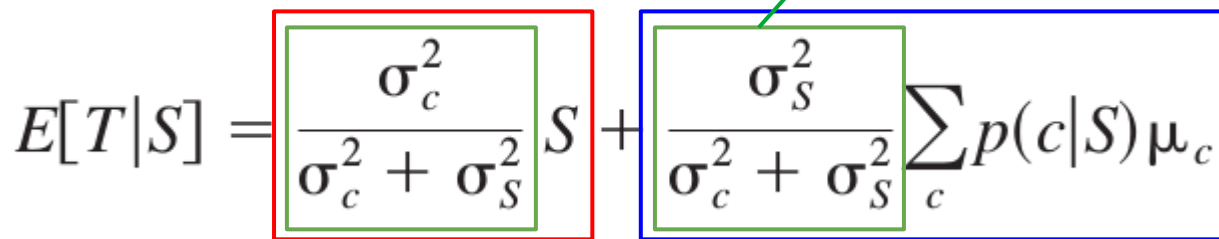
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This weight is larger when
the stimulus is **uncertain**!

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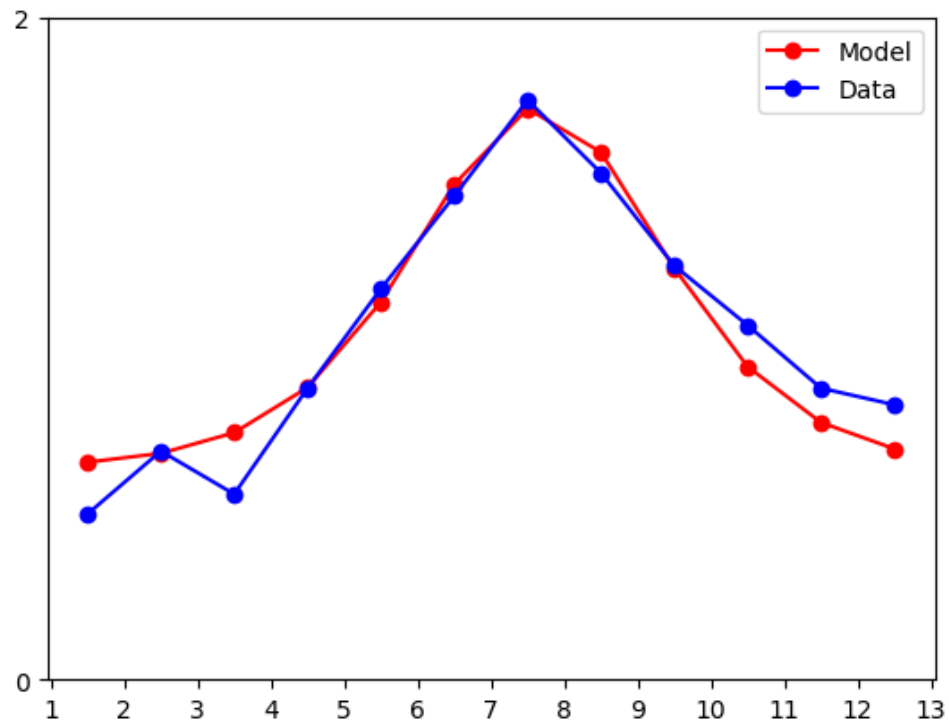
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The equation is presented with two main terms. The first term, $\frac{\sigma_c^2}{\sigma_c^2 + \sigma_s^2} S$, is enclosed in a red box. The second term, $\frac{\sigma_s^2}{\sigma_c^2 + \sigma_s^2} \sum_c p(c|S) \mu_c$, is enclosed in a blue box. Within each of these boxes, the fraction part is further enclosed in a green box. A green arrow points from the text 'This weight is larger when the stimulus is uncertain!' to the green box of the second term.

This weight is larger when
the stimulus is **uncertain!**

- 2 terms: **stimulus**, and **category means**
- Weighted by **certainty**
- **Prediction:** Increased **stimulus** uncertainty should strengthen **category** effect.

Model fit to data



Empirical data: the effect of stimulus noise

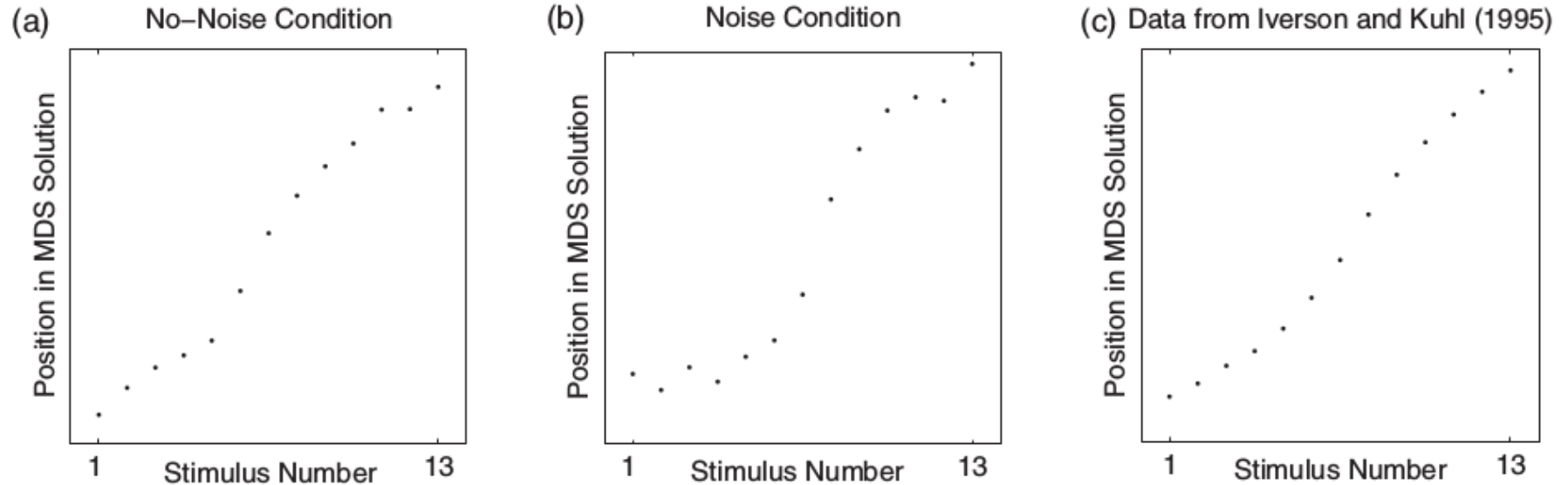


Figure 13. Perceptual maps for the (a) no-noise and (b) noise conditions obtained through multidimensional scaling (MDS). Data from (c) Iverson and Kuhl's (1995) multidimensional scaling experiment are shown for comparison.

Analytic form for mean of posterior distribution,
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Builds on a long tradition

Accounting in similar terms for effects of categories on:

- Memory for when an event occurred
- Memory for where a dot appeared

Builds on a long tradition

Accounting in similar terms for effects of categories on:

- Memory for when an event occurred
- Memory for where a dot appeared

Late 1980s, early 1990s: Bayesian cognitive science before it was fully “a thing”.

Memory for when an event occurred (Huttenlocher et al., 1988)

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- Participants were students at a university.

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- The university's academic year was divided into **3 temporal categories**: fall, winter, and spring quarters.

Memory for when an event occurred (Huttenlocher et al., 1988)

- Participants were students at a university.
- The university's academic year was divided into **3 temporal categories**: fall, winter, and spring quarters.
- At the **end of spring quarter**, participants were asked **when a particular film** was shown on campus.

What pattern of results would you predict, if participants were biasing their memories toward category prototypes, in line with the ideas we have just reviewed?

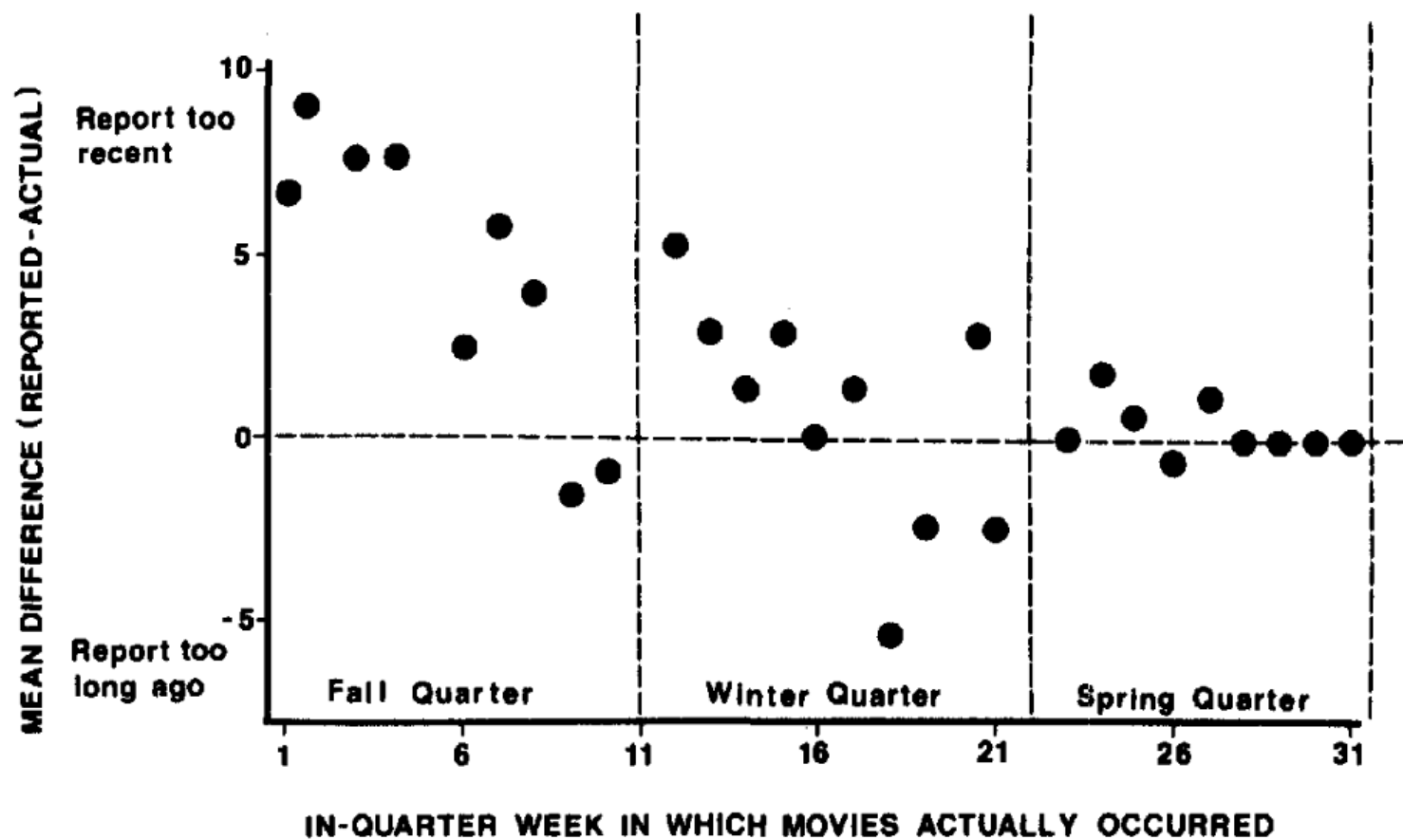


Figure 4. Plot of the mean differences (reported in-quarter weeks minus actual in-quarter weeks) by in-quarter week in which the movies actually occurred, for the 8-month group.

Memory for where a dot appeared (Huttenlocher et al., 1991)

Memory for where a dot appeared (Huttenlocher et al., 1991)

- A dot appeared in a circle, then vanished.

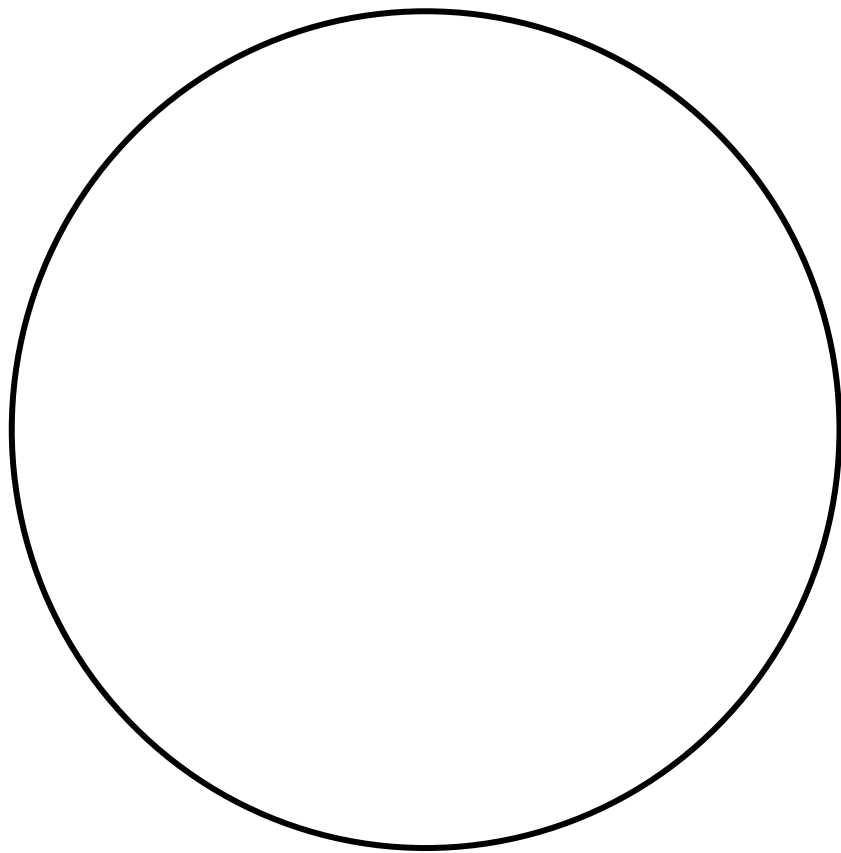
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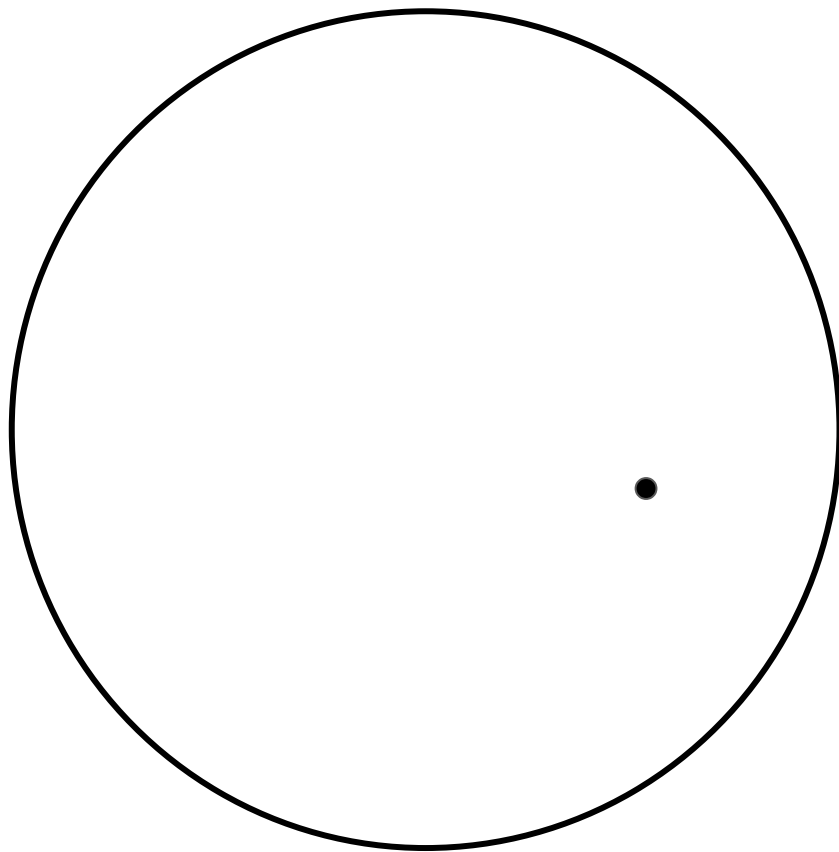
- A dot appeared in a circle, then vanished.
- Participants were asked to indicate where the dot had appeared.

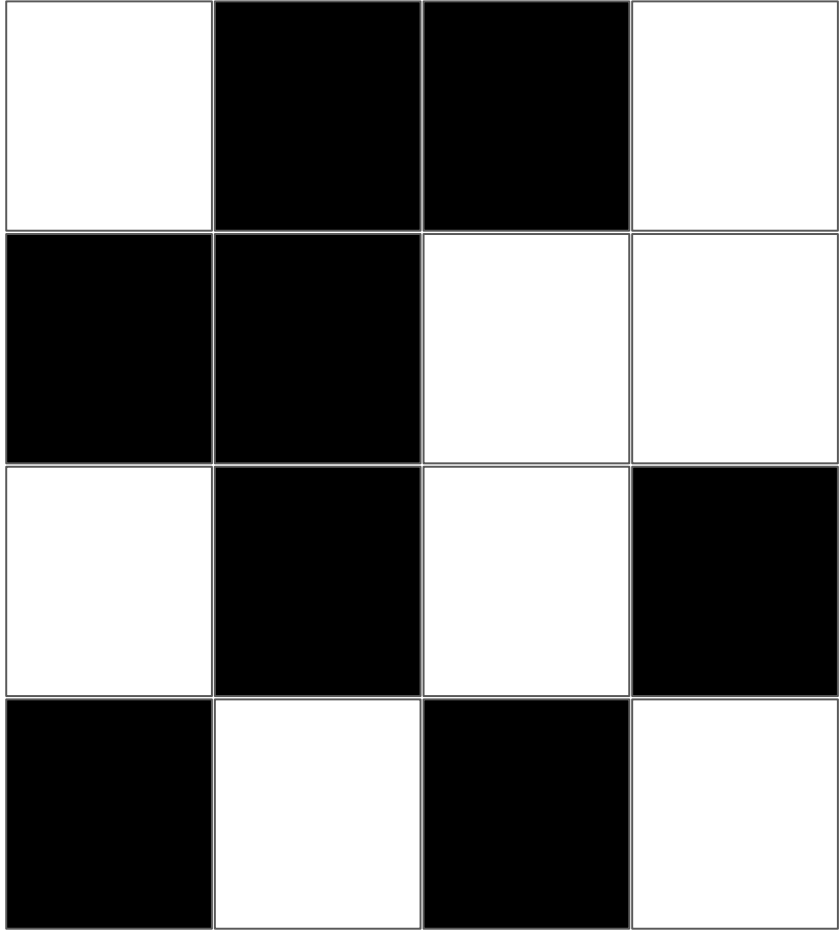
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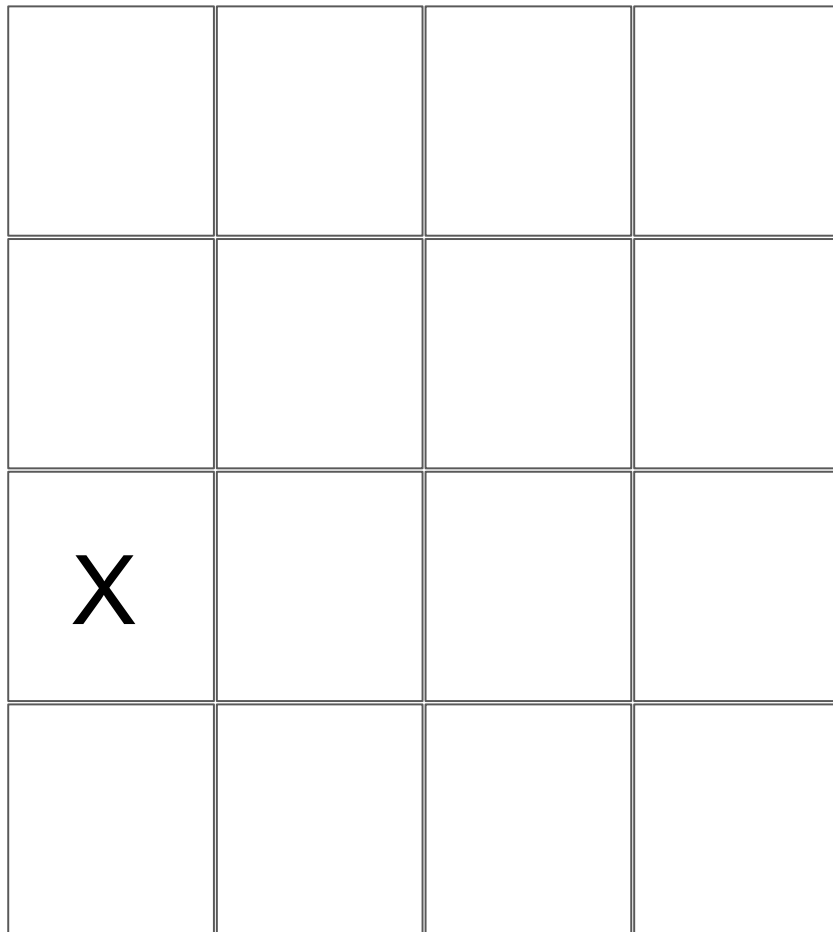
- A dot appeared in a circle, then vanished.
- Participants were asked to indicate where the dot had appeared.
- This was done either with or without a visual interference task.



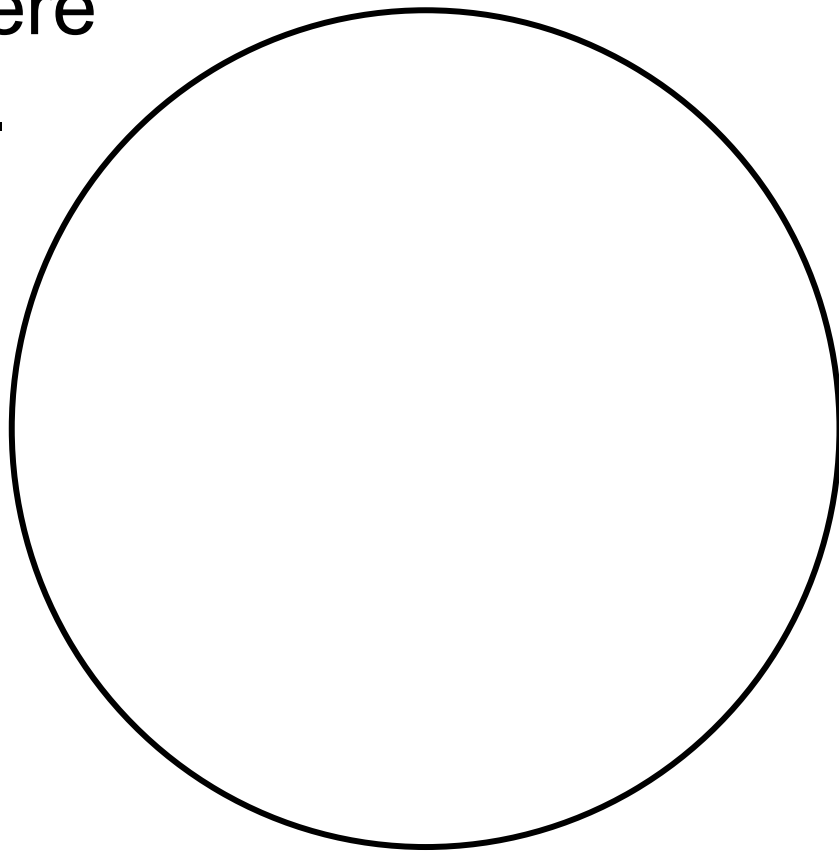




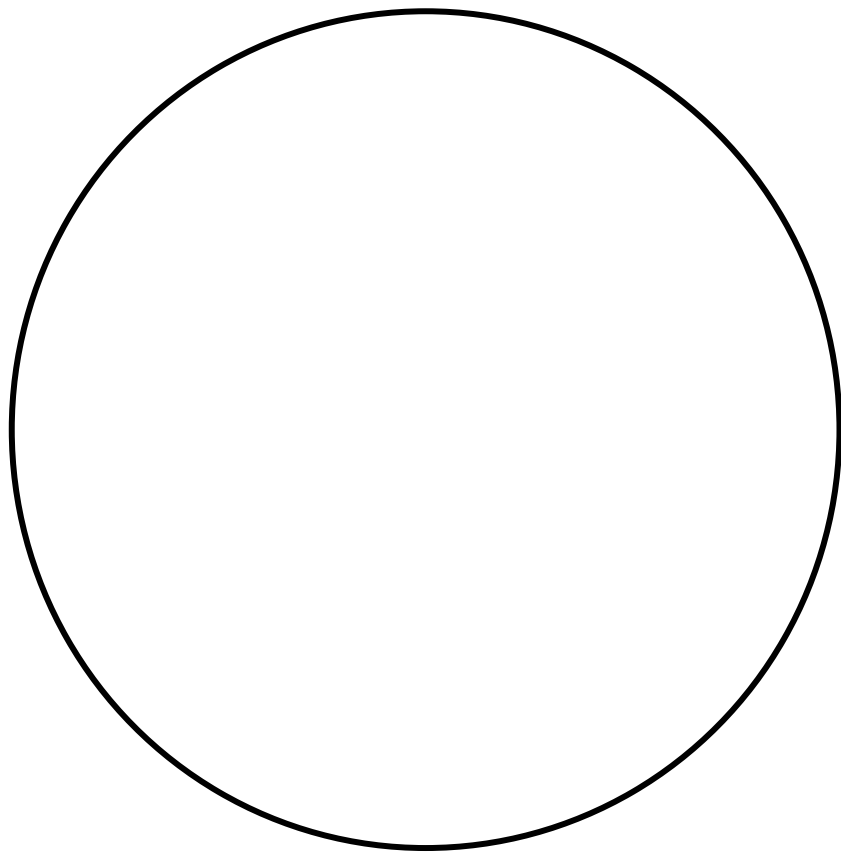
Was the
square
marked X
black or
white?

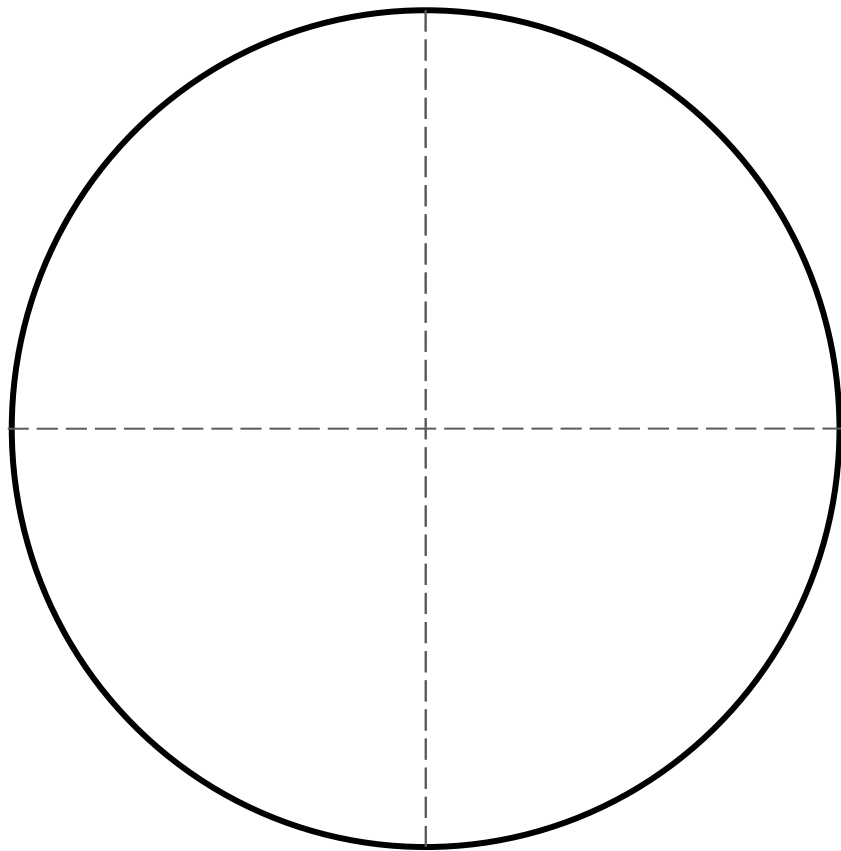


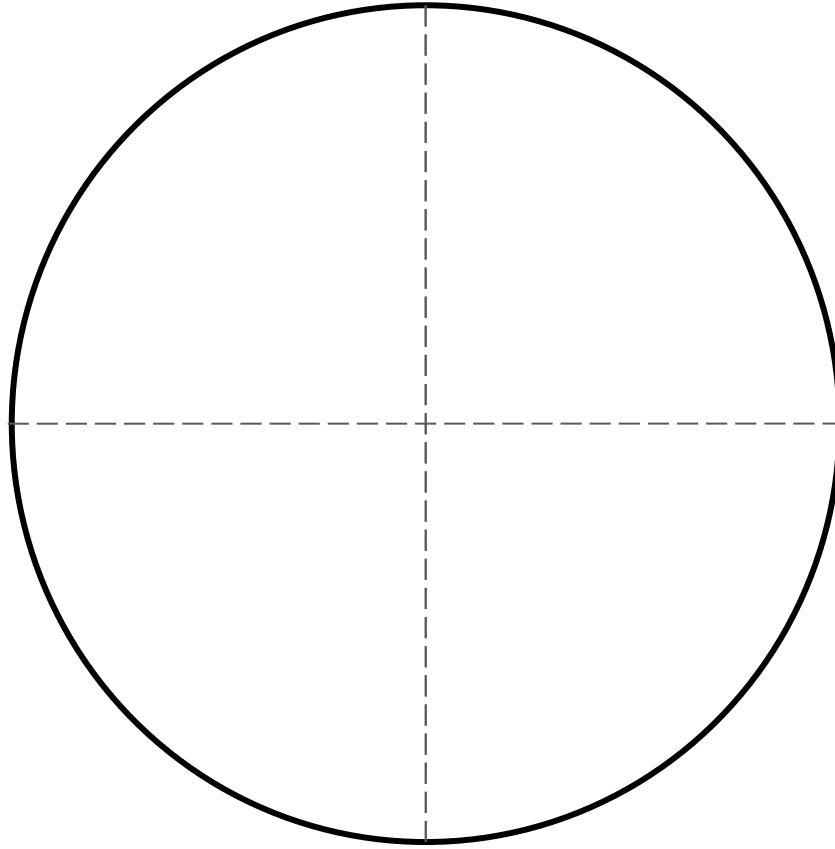
Indicate where
the dot was.



What pattern of results would you predict, if participants were biasing their memories toward category prototypes, in line with the ideas we have just reviewed?







Greater bias
in interference
condition.

The Sapir-Whorf hypothesis



The Sapir-Whorf hypothesis

Our thoughts are shaped by our native language, and speakers of different languages therefore think differently.



The Sapir-Whorf hypothesis

Controversial

The Sapir-Whorf hypothesis

Controversial

- Appears to deny the possibility of a universal groundwork for cognition.

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- Appears to deny the possibility of a universal groundwork for cognition.
- Evidence for it has not always replicated.

The Sapir-Whorf hypothesis

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- Appears to deny the possibility of a universal groundwork for cognition.
- Evidence for it has not always replicated.

Claim: Viewing this hypothesis through the lens of **probabilistic inference** has the potential to resolve **both** issues.

Proposal

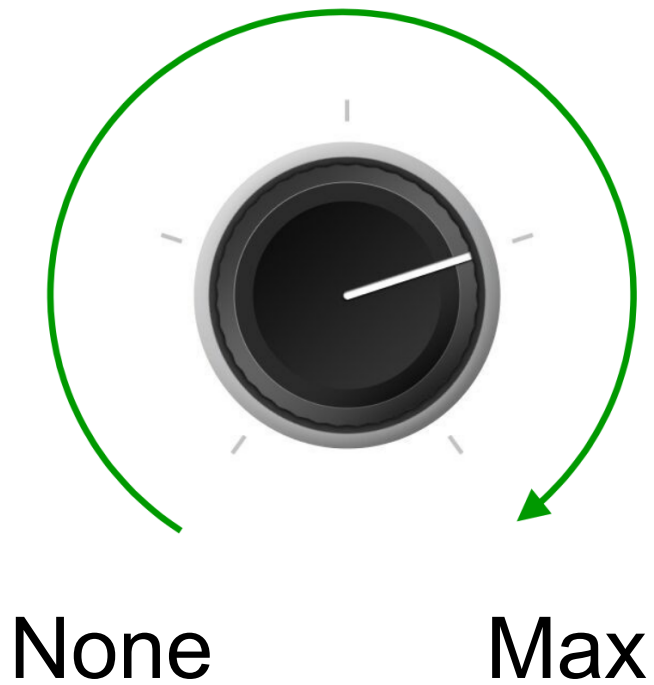
Testing the proposal

Proposal

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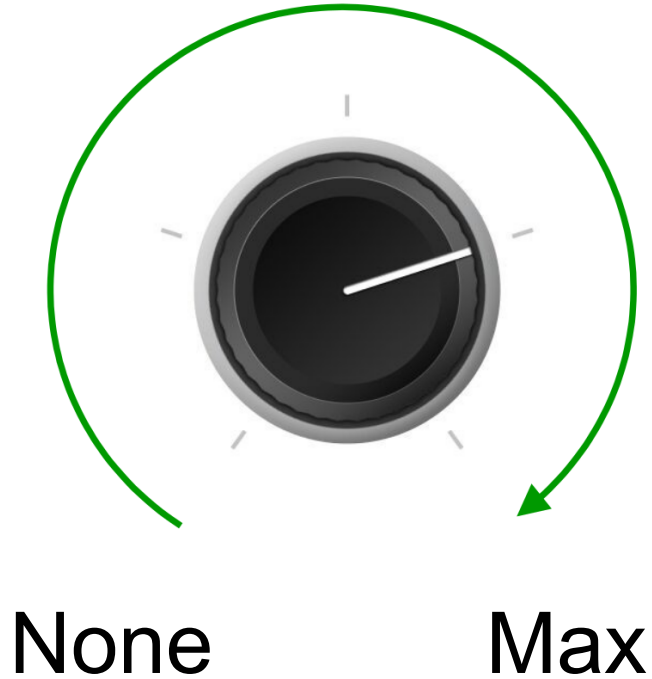


Whorf



Whorf

Uncertainty



Whorf

Category adjustment model

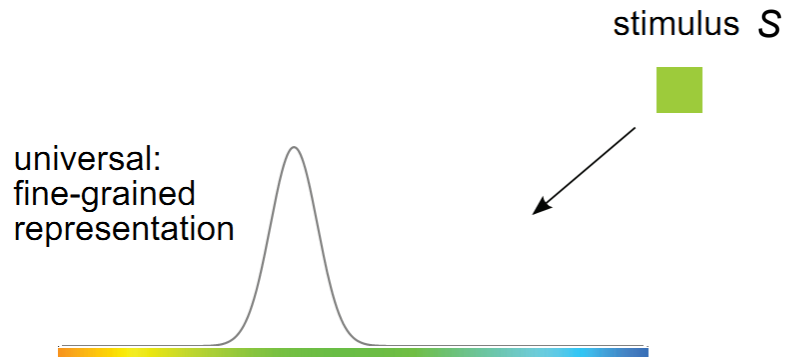
Huttenlocher et al., 1991; Bae et al., 2015; Persaud & Hemmer, 2014

stimulus S



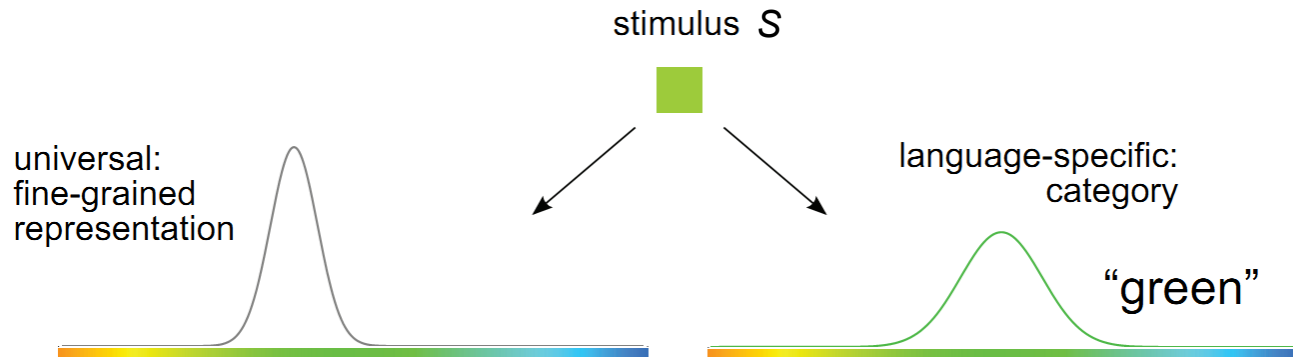
Category adjustment model

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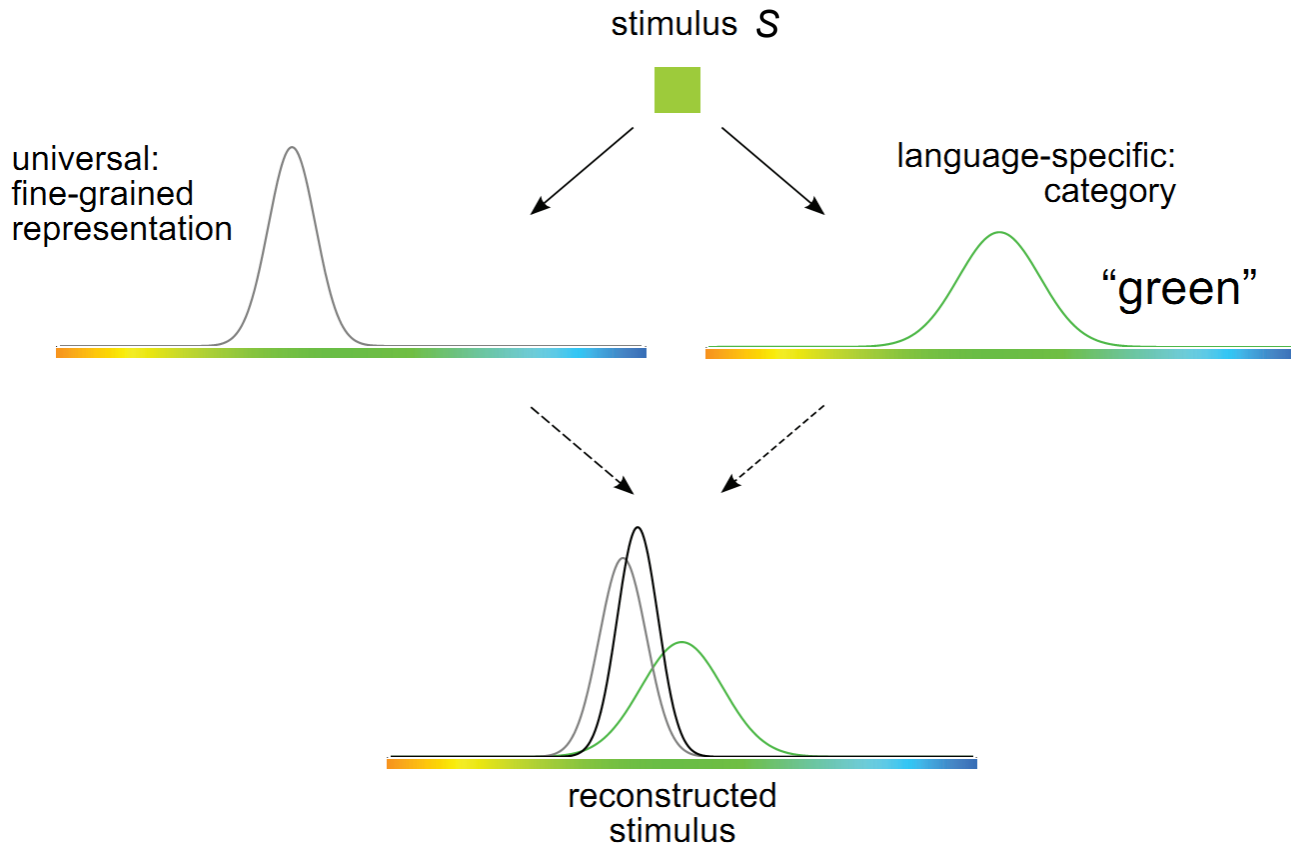
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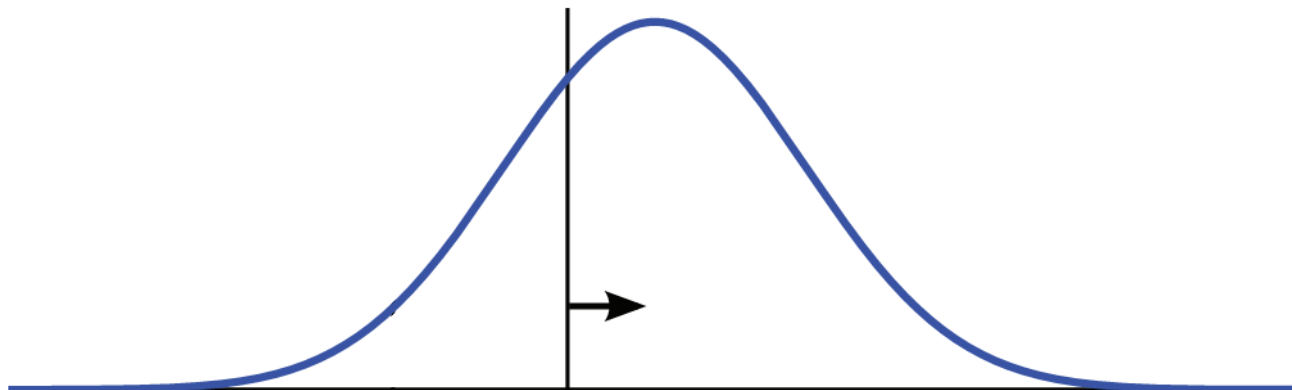
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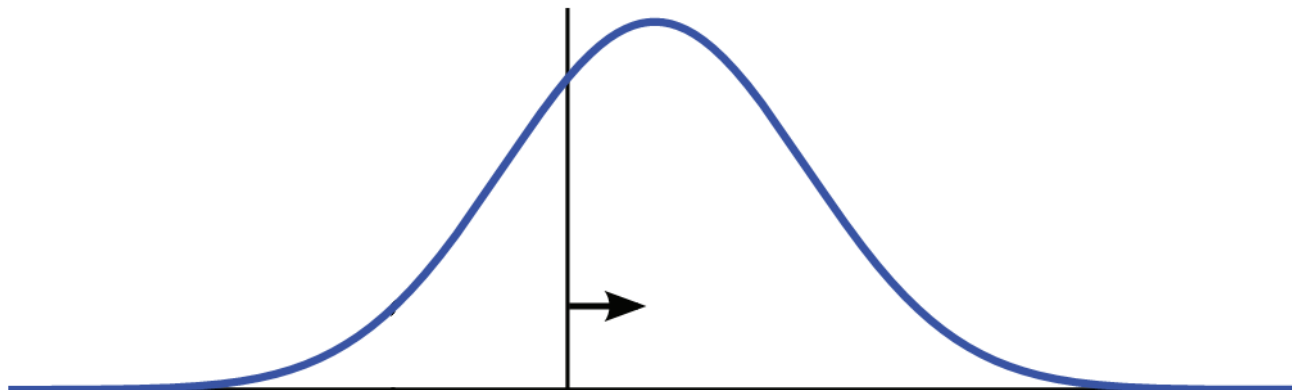
Bias toward category prototype



Category adjustment model

Huttenlocher et al., 1991; Bae et al., 2015; Persaud & Hemmer, 2014

Bias toward category prototype
More bias with more uncertainty



Category adjustment model

Huttenlocher et al., 1991; Bae et al., 2015; Persaud & Hemmer, 2014

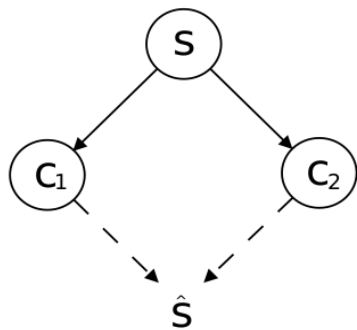


Figure 1. A stimulus S produces two cues c_1, c_2 . From these cues, we obtain an estimate $\hat{S}$ of the original stimulus.

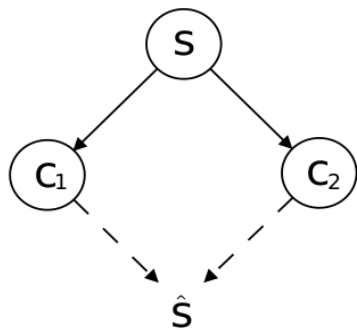
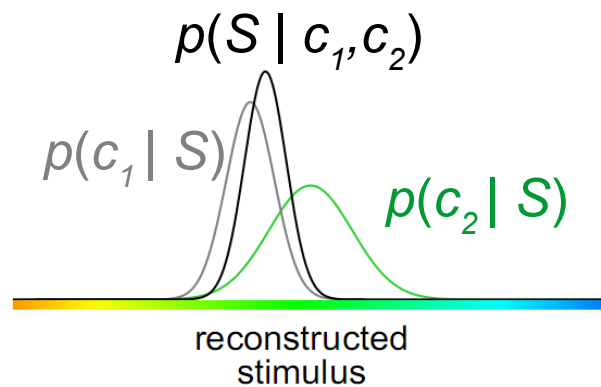


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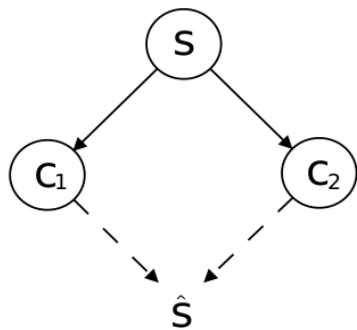
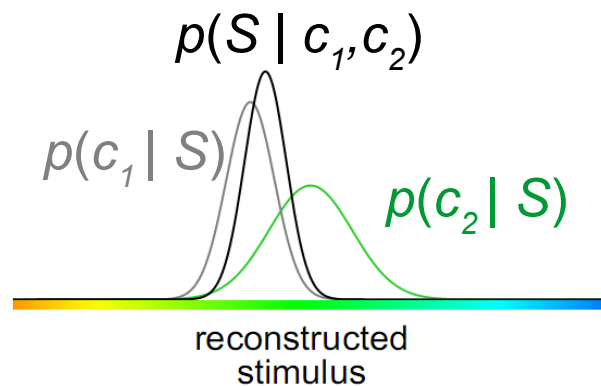


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$$\begin{aligned}
 p(S|c_1, c_2) &\propto p(c_1, c_2|S)p(S) && \text{[Bayes' rule]} && (1) \\
 &= p(c_1|c_2, S)p(c_2|S)p(S) && && (2) \\
 &= p(c_1|S)p(c_2|S)p(S) && \text{[because } c_1 \text{ is independent of } c_2 \text{ given } S] && (3) \\
 &\propto p(c_1|S)p(c_2|S) && \text{[because } p(S) \text{ is uniform]} && (4) \\
 &= \mathcal{N}(\mu_1, \sigma_1^2) \mathcal{N}(\mu_2, \sigma_2^2) && \text{[because } c_1, c_2 \text{ are normally distributed]} && (5)
 \end{aligned}$$

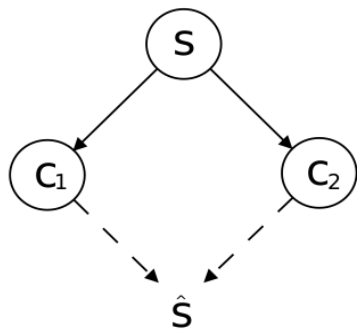
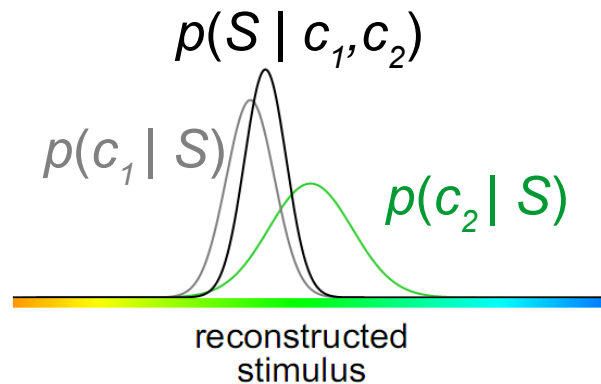


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$$p(S|c_1, c_2) \propto p(c_1, c_2|S)p(S) \quad [\text{Bayes' rule}] \quad (1)$$

$$= p(c_1|c_2, S)p(c_2|S)p(S) \quad (2)$$

$$= p(c_1|S)p(c_2|S)p(S) \quad [\text{because } c_1 \text{ is independent of } c_2 \text{ given } S] \quad (3)$$

$$\propto p(c_1|S)p(c_2|S) \quad [\text{because } p(S) \text{ is uniform}] \quad (4)$$

$$= \mathcal{N}(\mu_1, \sigma_1^2) \mathcal{N}(\mu_2, \sigma_2^2) \quad [\text{because } c_1, c_2 \text{ are normally distributed}] \quad (5)$$

The product of these two normal distributions is itself proportional to a normal distribution with:

$$\mu_{12} = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} \mu_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \mu_2 \quad \text{and} \quad \sigma_{12}^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} \quad (6)$$

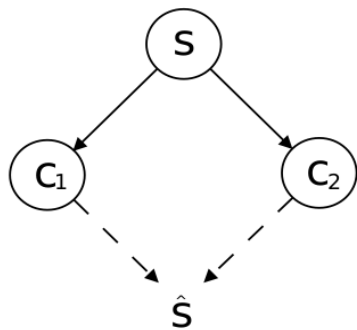
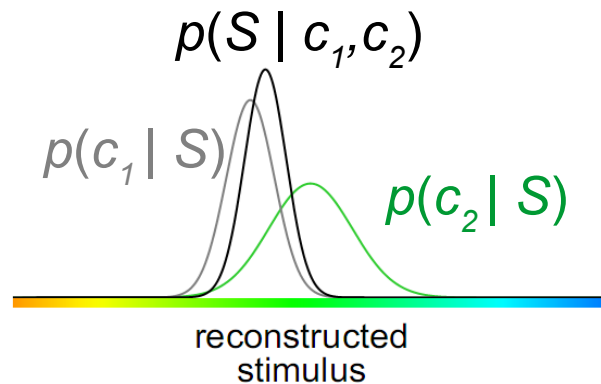


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This weight is larger when the stimulus is uncertain!

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$$= p(c_1|c_2, S)p(c_2|S)p(S) \quad (2)$$

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Ideas with broad application

Ideas with broad application

Memory for when an event occurred, affected by temporal categories. Huttenlocher et al. 1988

Ideas with broad application

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Proposal

Testing the proposal

Predictions

Predictions

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Predictions

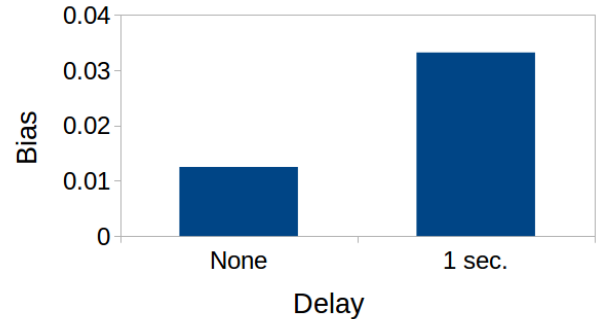
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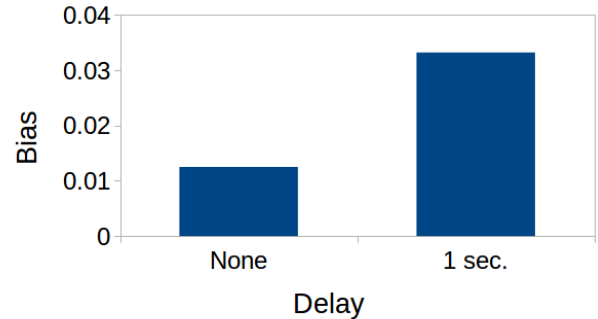
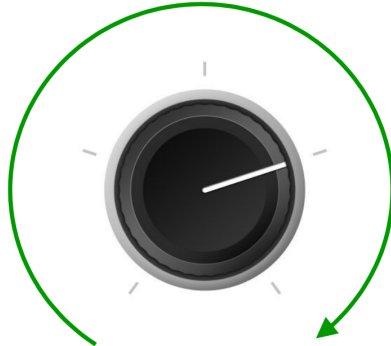


Cibelli et al., 2016

PLOS ONE

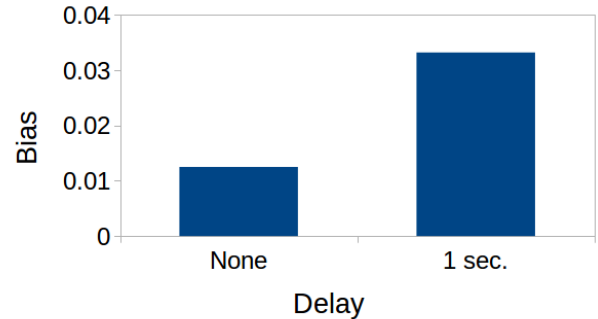
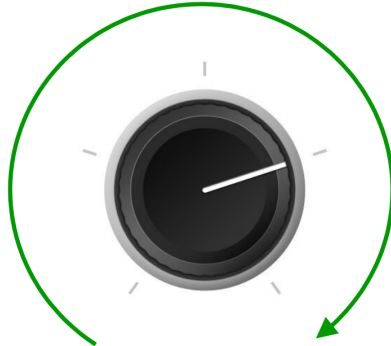
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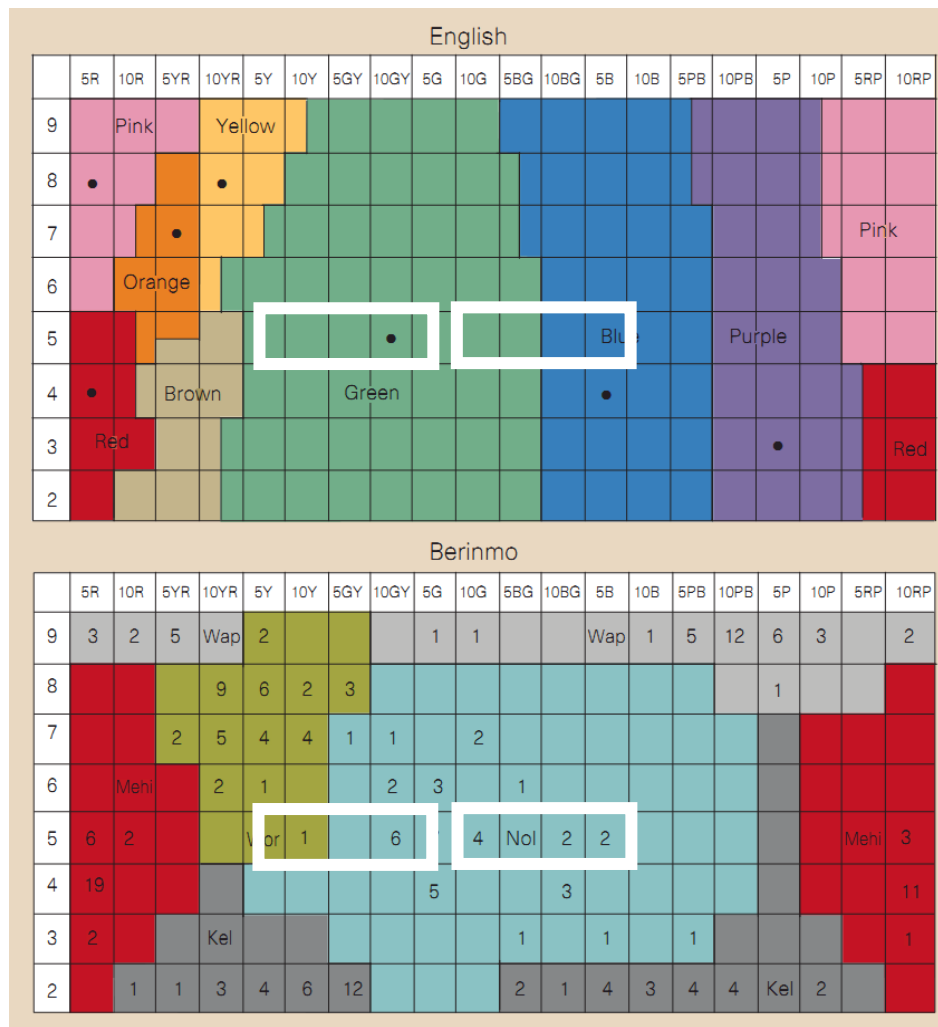
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English vs. Berinmo

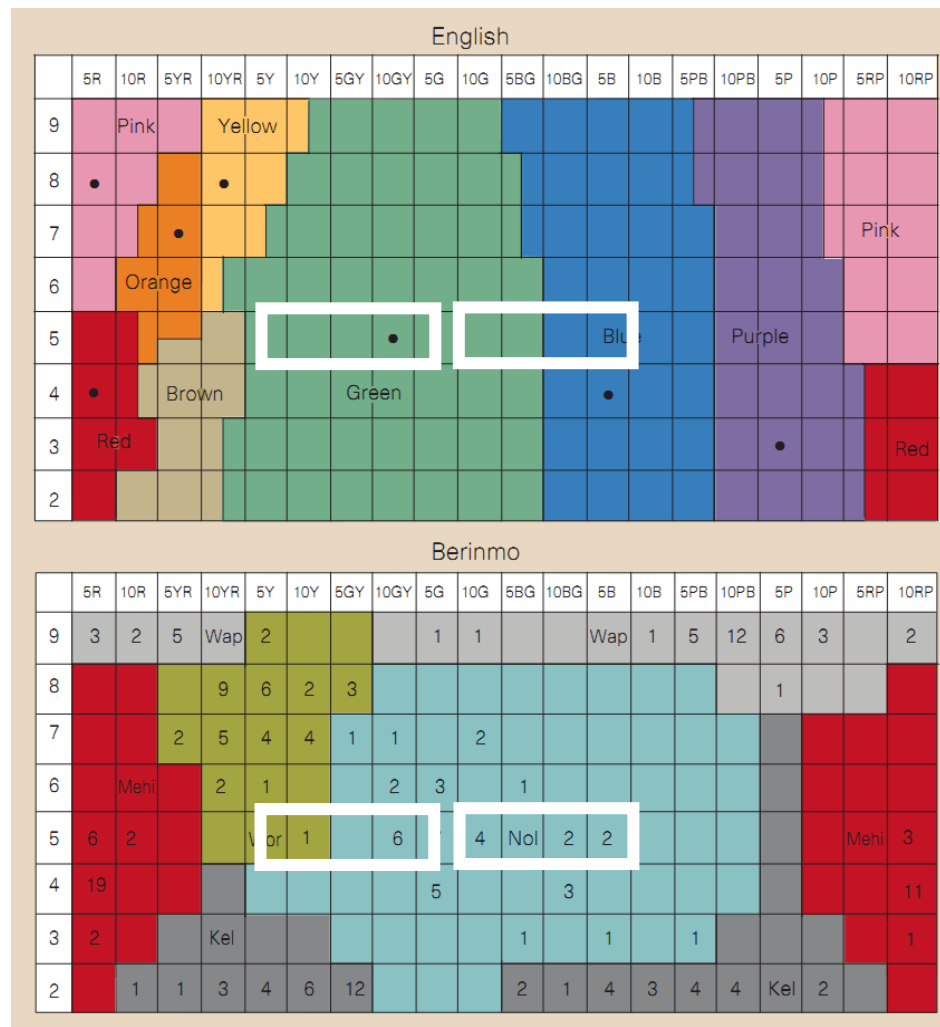


Davidoff et al., 1999, *Nature*

English vs. Berinmo

- Color discrimination from memory

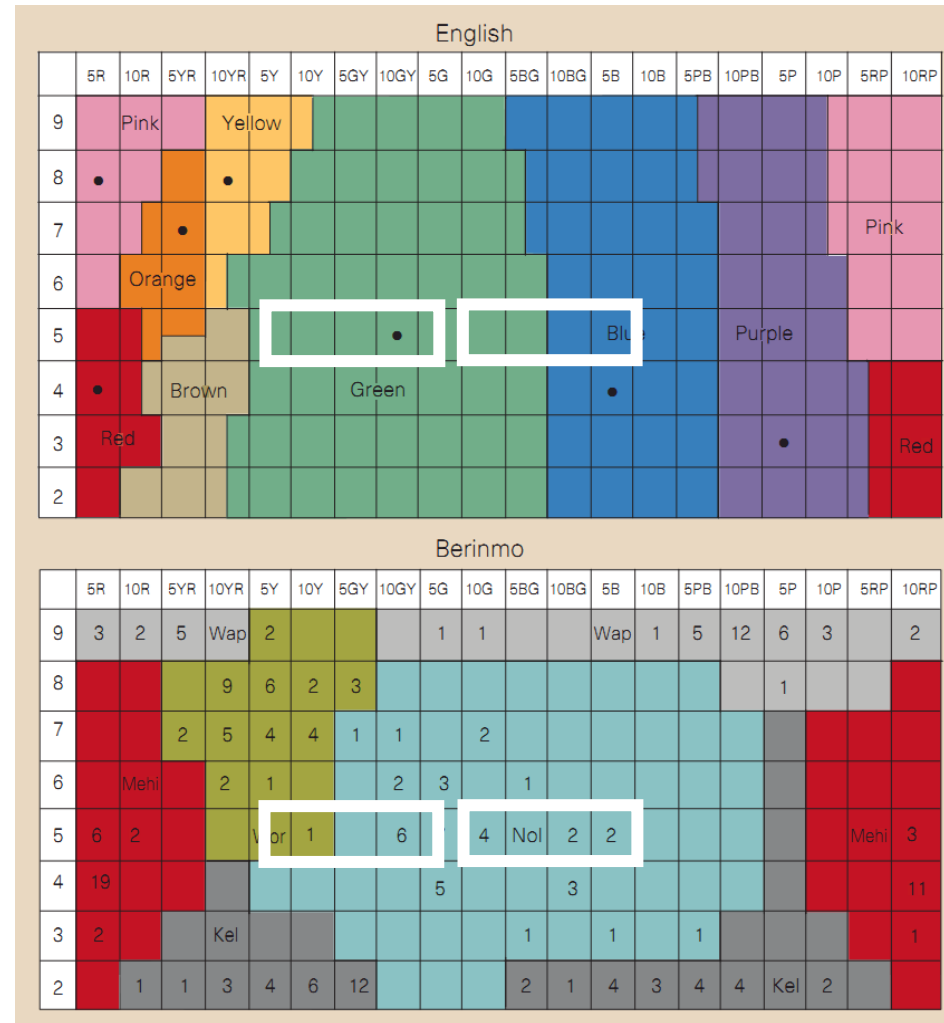
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English vs. Berinmo

- Color discrimination from memory
- 2 AFC task

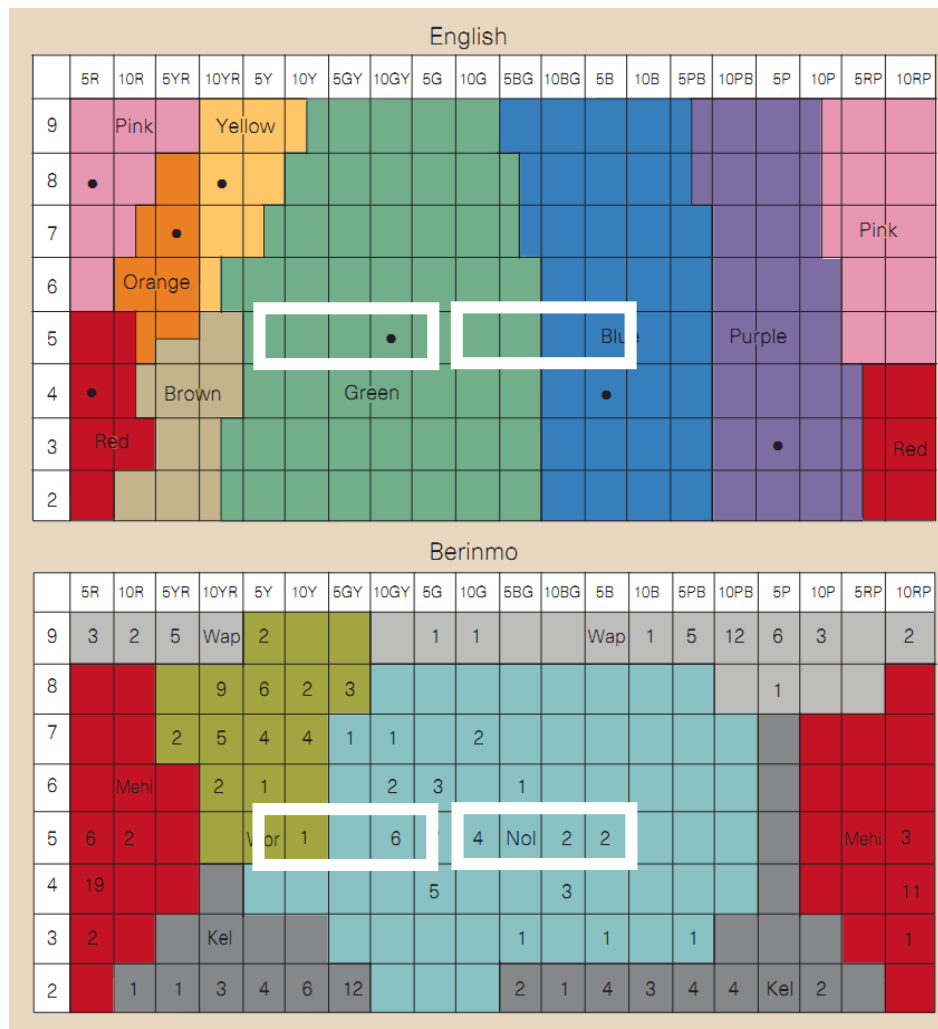
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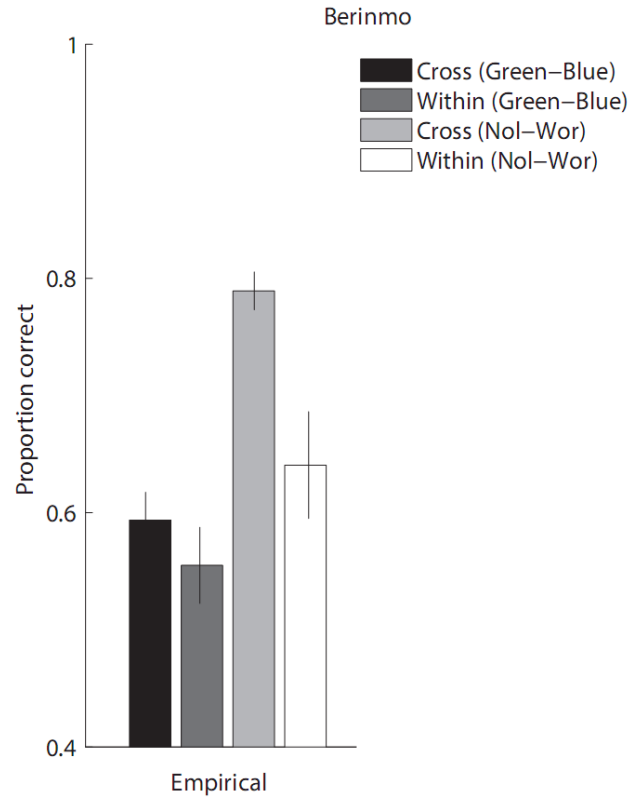


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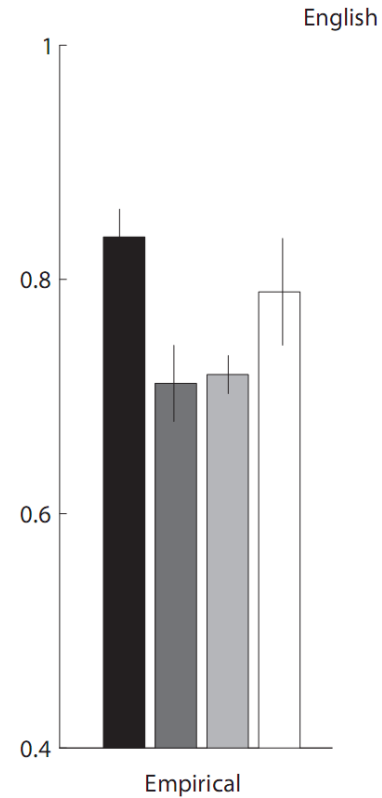
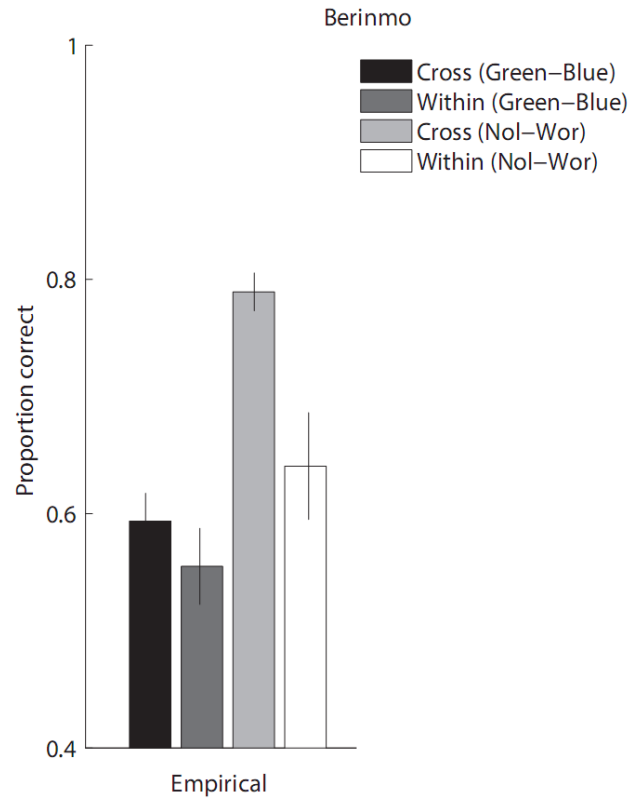
- Color discrimination from memory
- 2 AFC task
- Cross-category vs. within-category

Davidoff et al., 1999, *Nature*

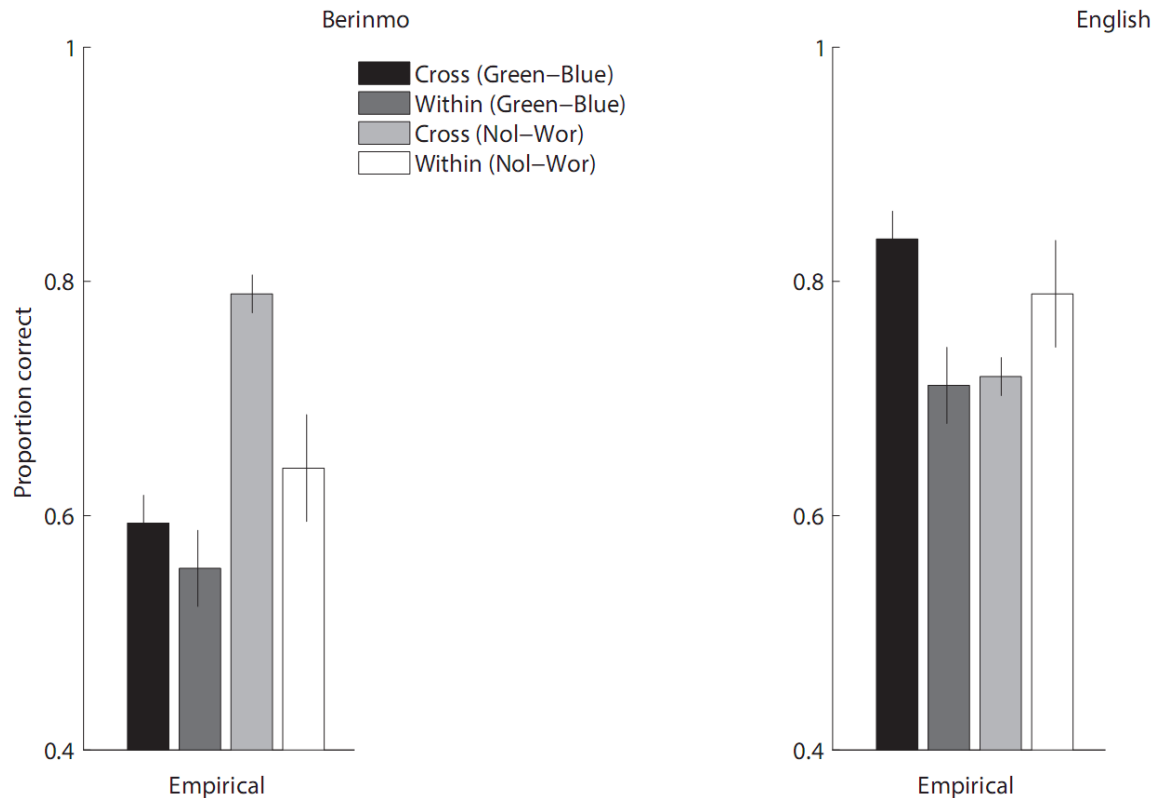




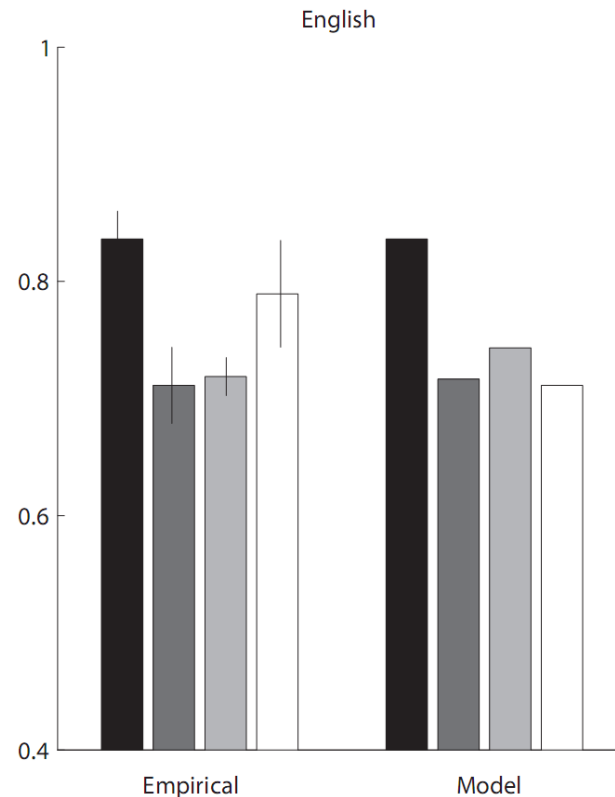
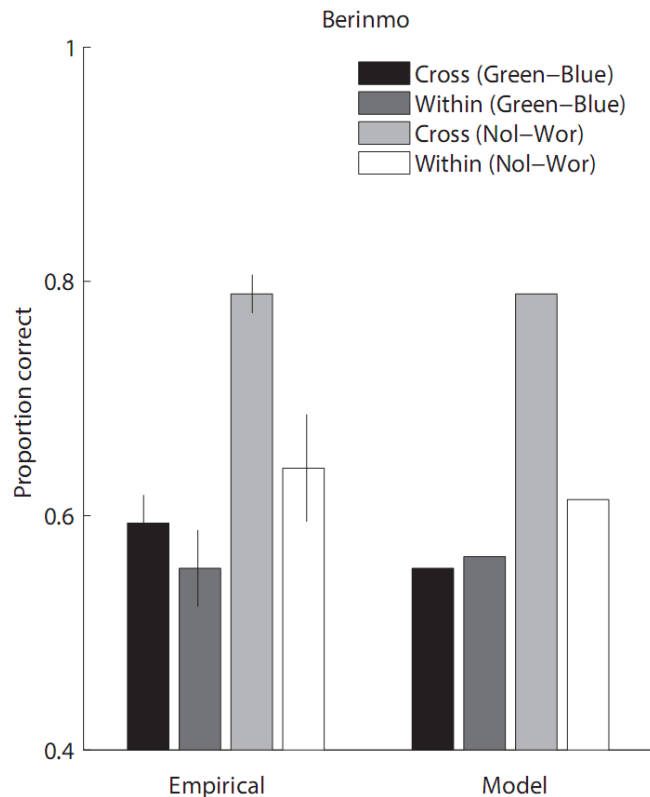
Data from Roberson et al. (2000)



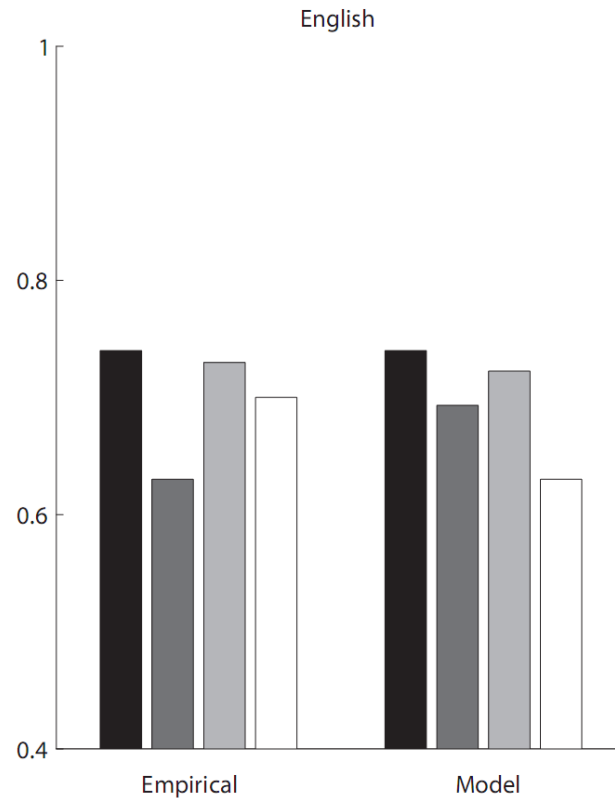
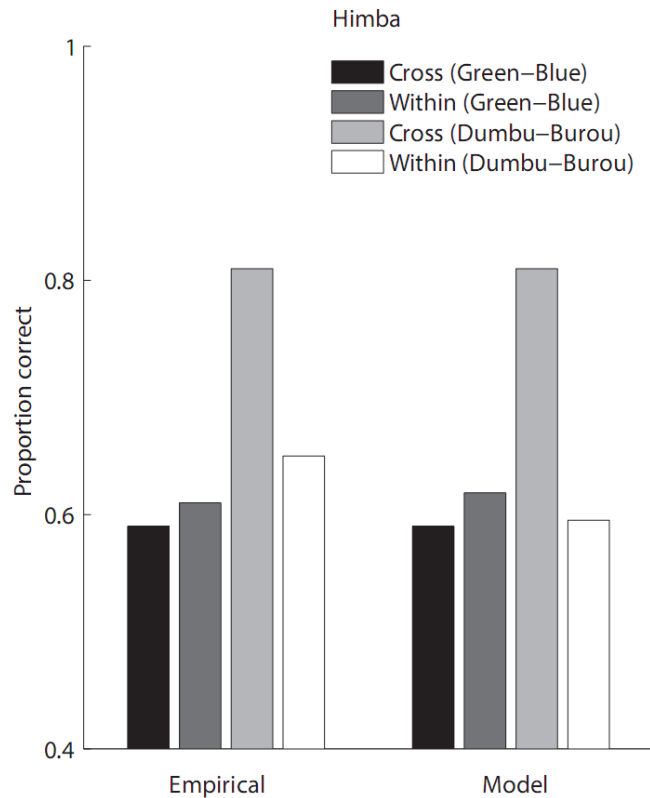
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One model per language
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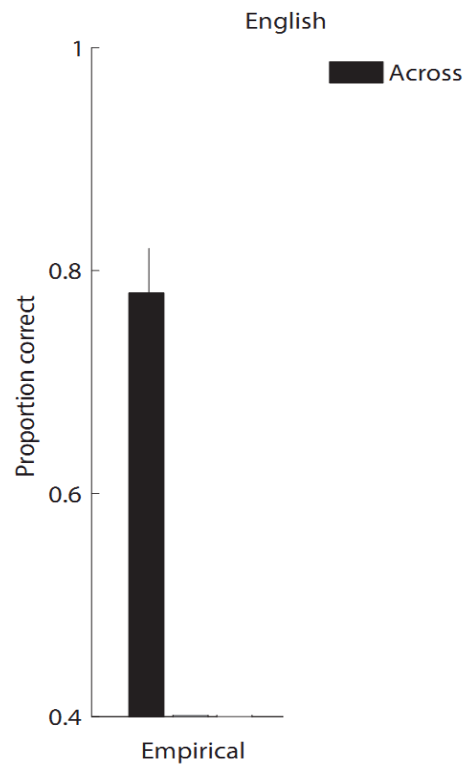
One model per language
Data from Roberson et al. (2005)

Discrimination data are fit best by **native-language** models

Berinmo-English comparison		
	Berinmo model	English model
Berinmo discrimination data	0.0017	0.0080
English discrimination data	0.0019	0.0006
Himba-English comparison		
	Himba model	English model
Himba discrimination data	0.0022	0.0039
English discrimination data	0.0029	0.0008

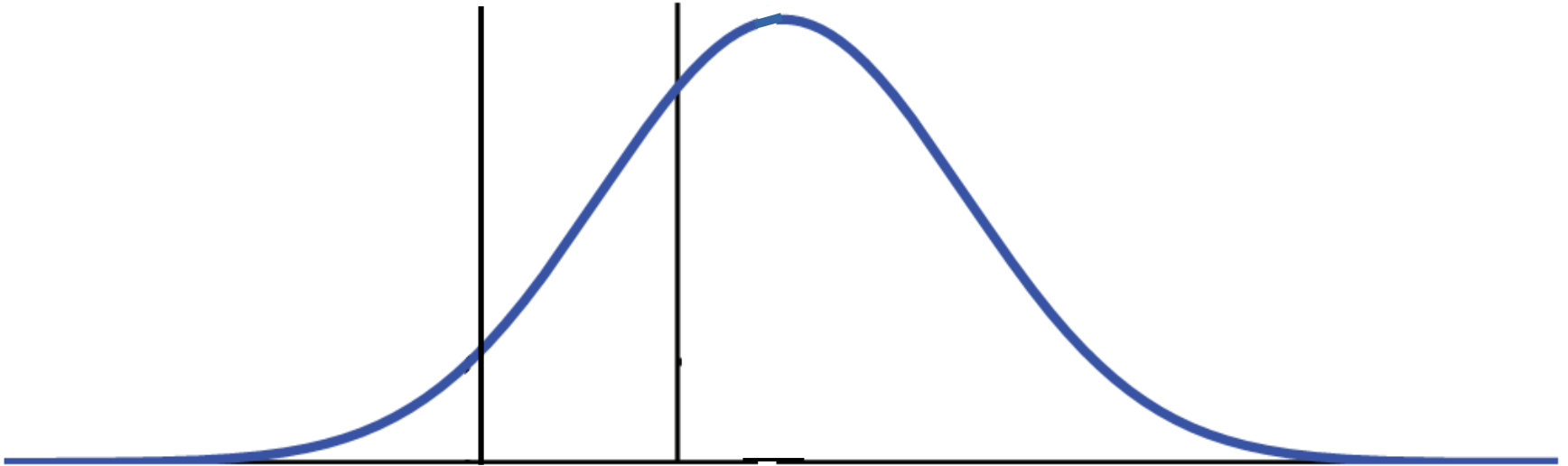
One last observation

One last observation



Data from Hanley and Roberson (2011)

Within-category

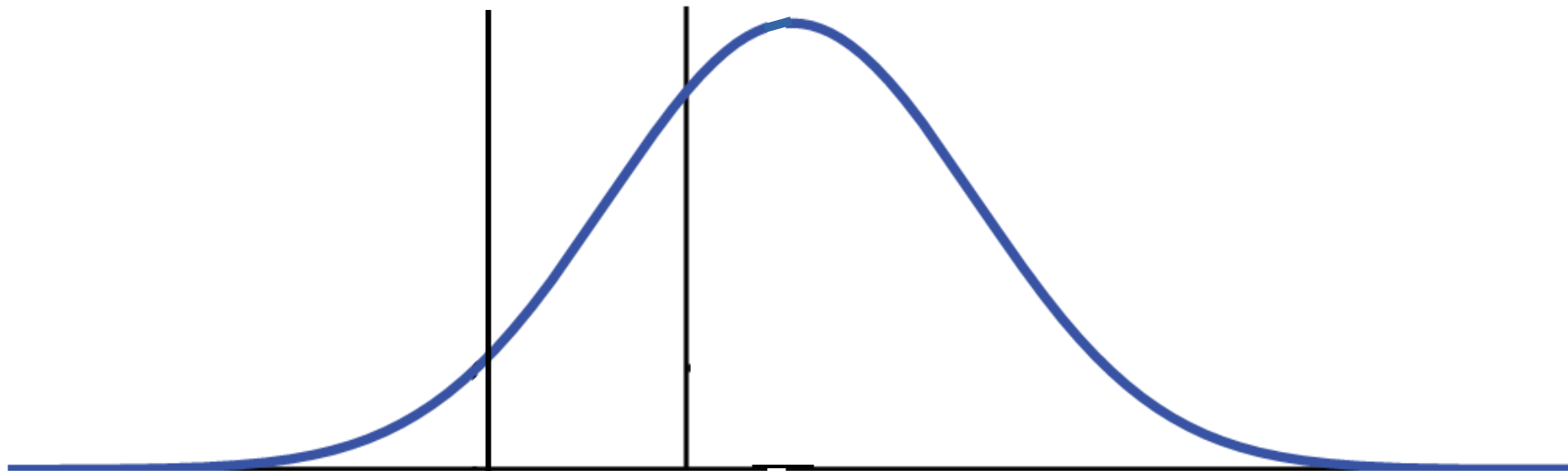


Within-category

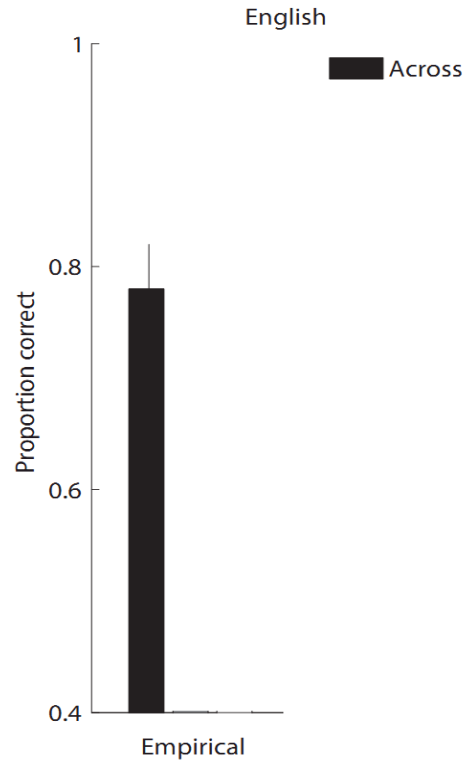


PE

GE

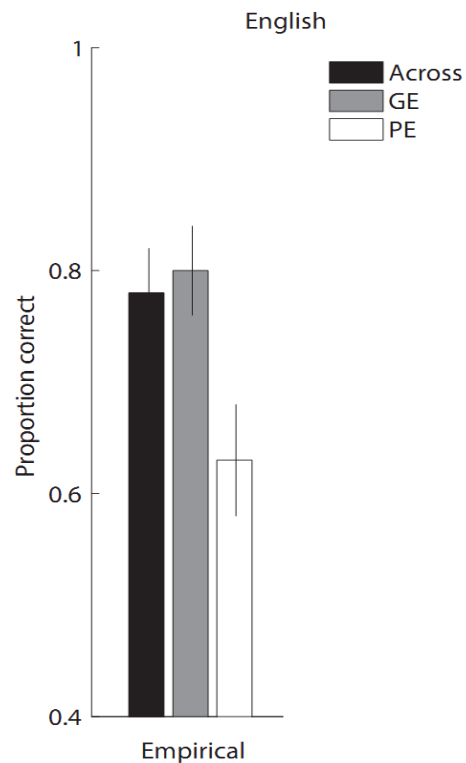


Within-category effects



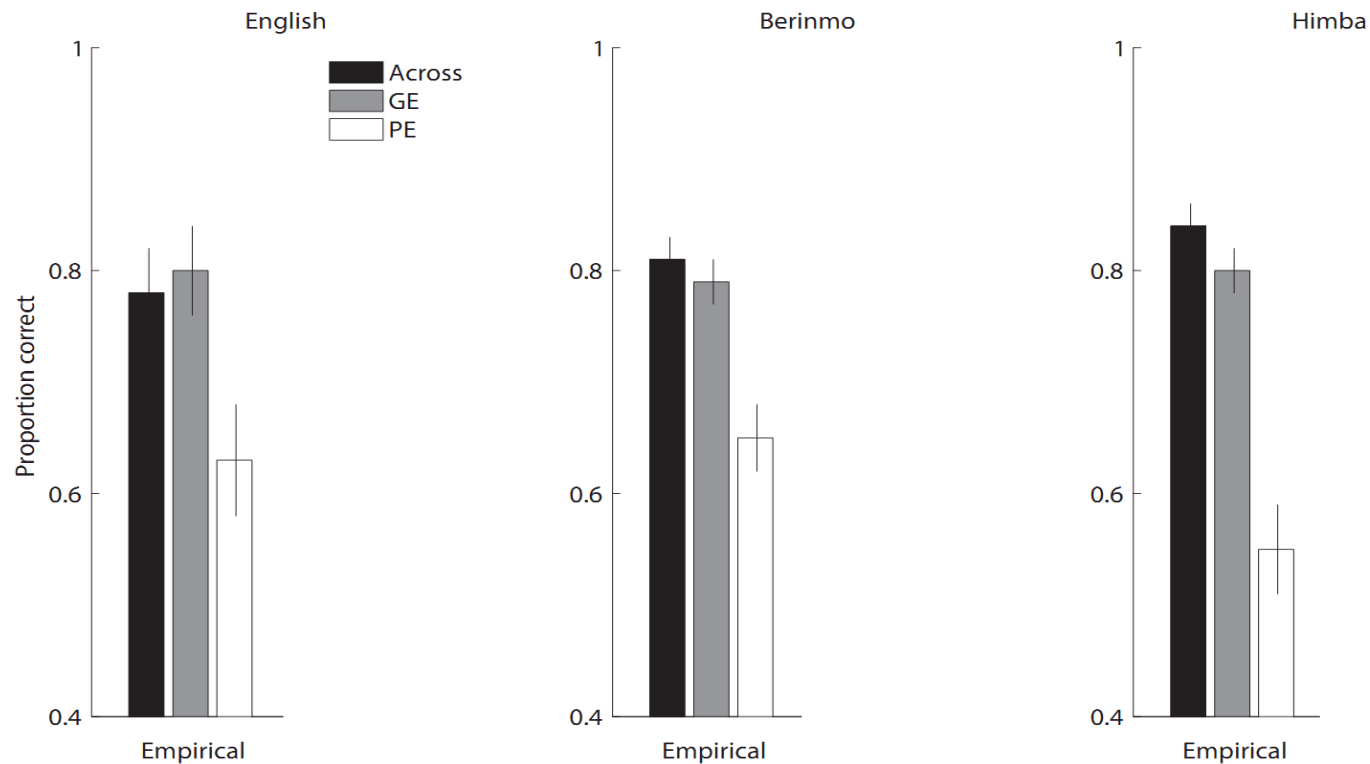
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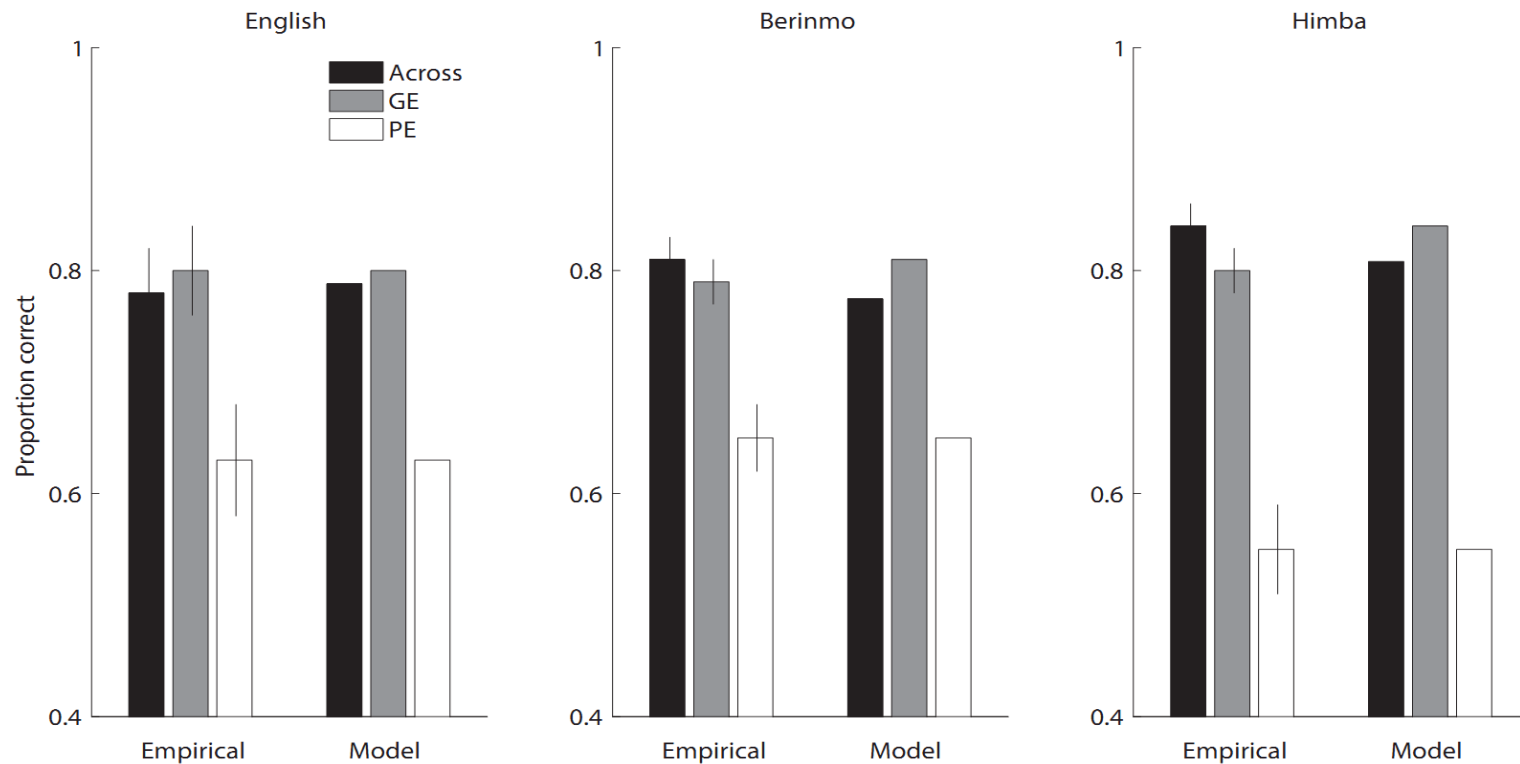
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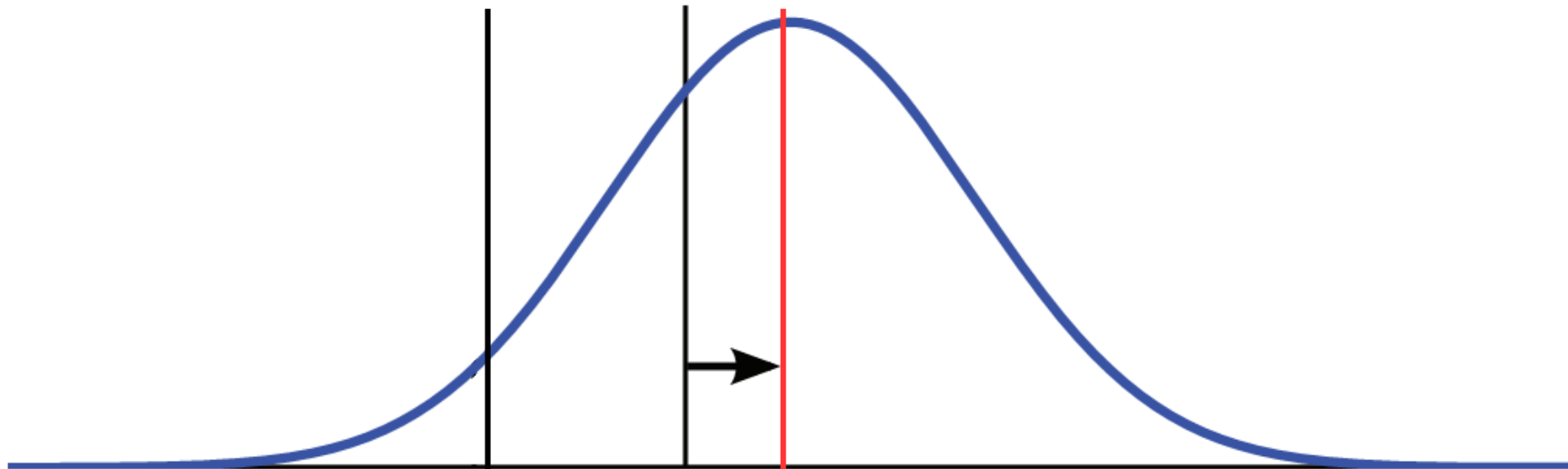
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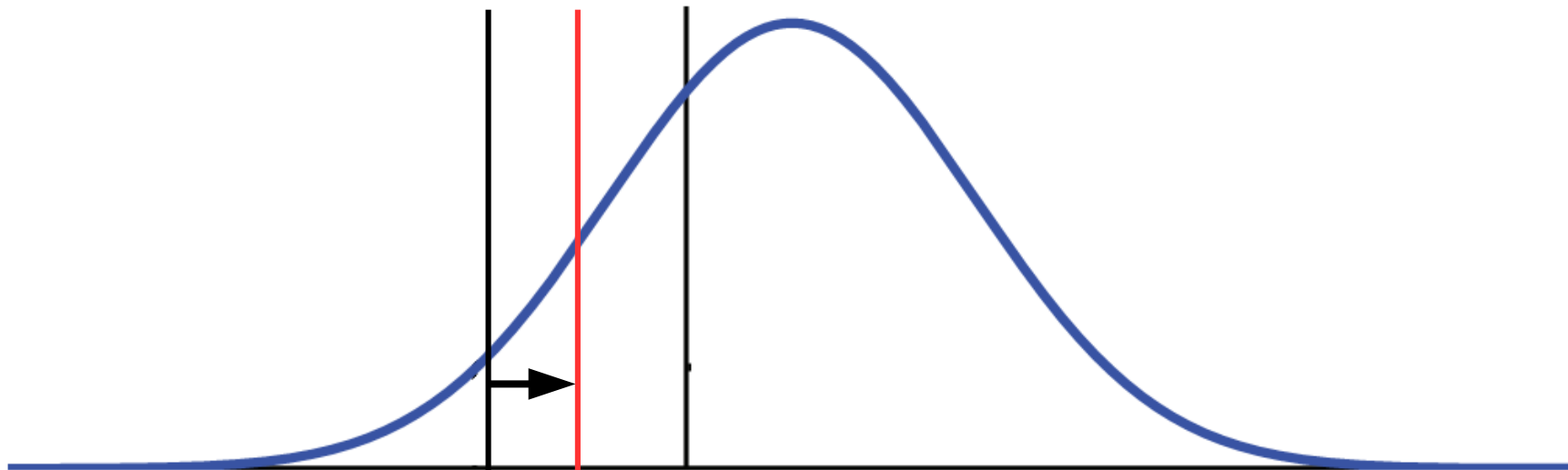


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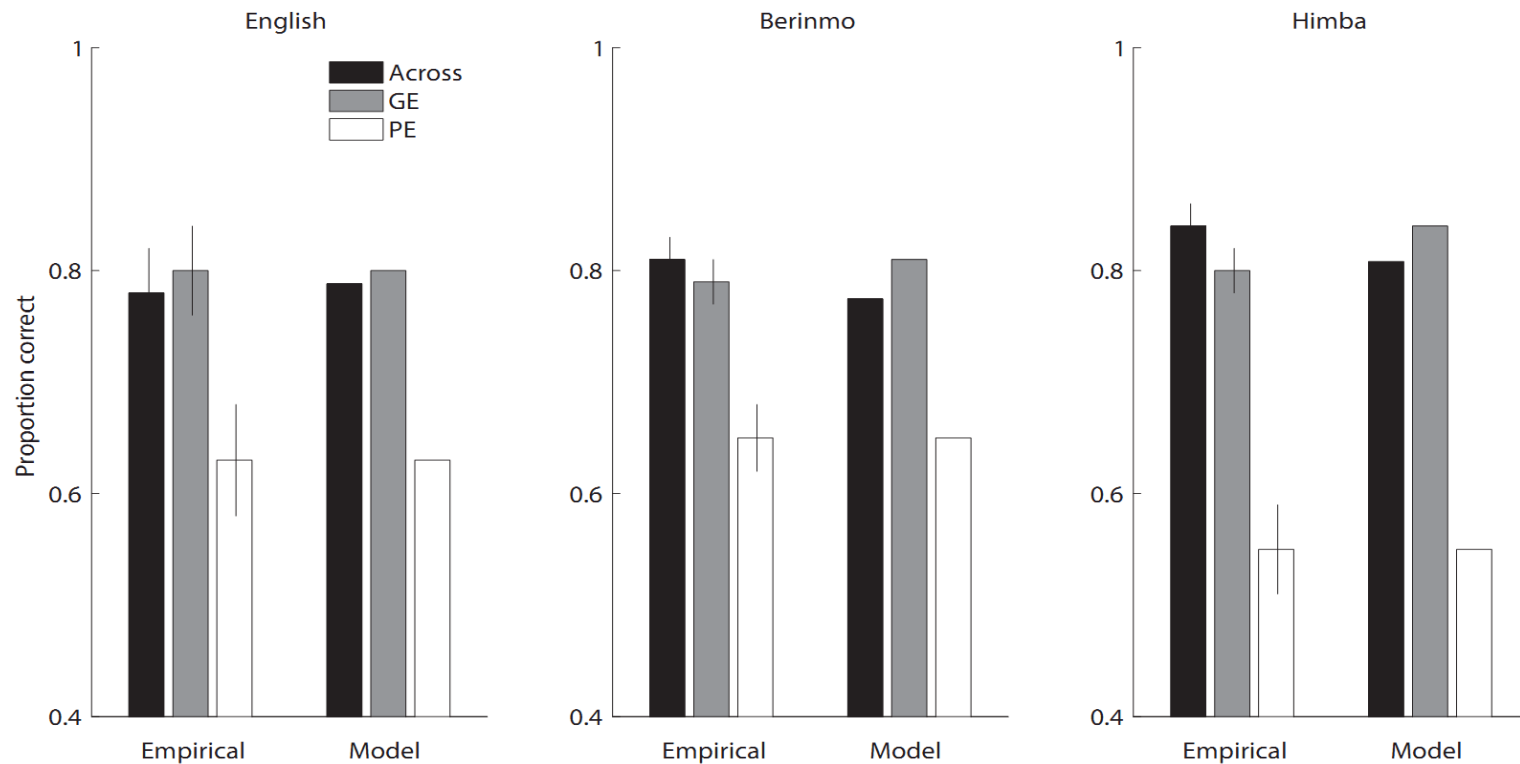
GE



PE



Within-category effects



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Predictions

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- Bias should be greater with delay than without.
- Model should account for existing cross-language data on color memory.

Cross-language differences in color memory are consistent with **reconstruction bias** under **uncertainty**, guided by **native-language categories**, in a **universal** color space.

Conclusions: Sapir-Whorf hypothesis

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Viewing the Sapir-Whorf hypothesis in
terms of probabilistic inference:

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- May resolve two major sources of controversy concerning this hypothesis

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Viewing the Sapir-Whorf hypothesis in terms of probabilistic inference:

- May resolve two major sources of controversy concerning this hypothesis
- Links this hypothesis to independent principles and phenomena

The general idea

Memory and perception as probabilistic **inference**, guided by **categories**, and modulated by stimulus **uncertainty**.

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Memory and perception as probabilistic **inference**, guided by **categories**, and modulated by stimulus **uncertainty**.

- Perceptual magnet, vowel perception
- Memory for events, dot locations, ...
- Sapir-Whorf hypothesis

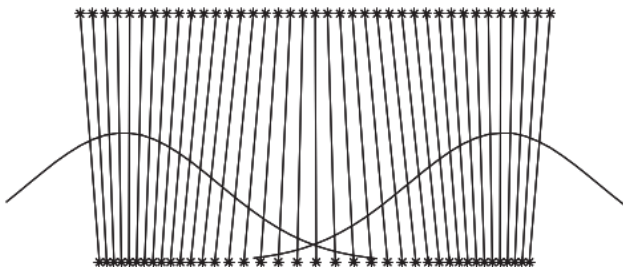


This lecture is dedicated to the memory of
Janelle Huttenlocher (1932-2016).

Questions? Comments?

(b)

Actual Stimulus



Perceived Stimulus

$$E[T|S] = \frac{\sigma_c^2}{\sigma_c^2 + \sigma_S^2} S + \frac{\sigma_S^2}{\sigma_c^2 + \sigma_S^2} \sum_c p(c|S) \mu_c$$

